



Metal-Organic Frameworks (MOFs)

- from basic research to potential applications

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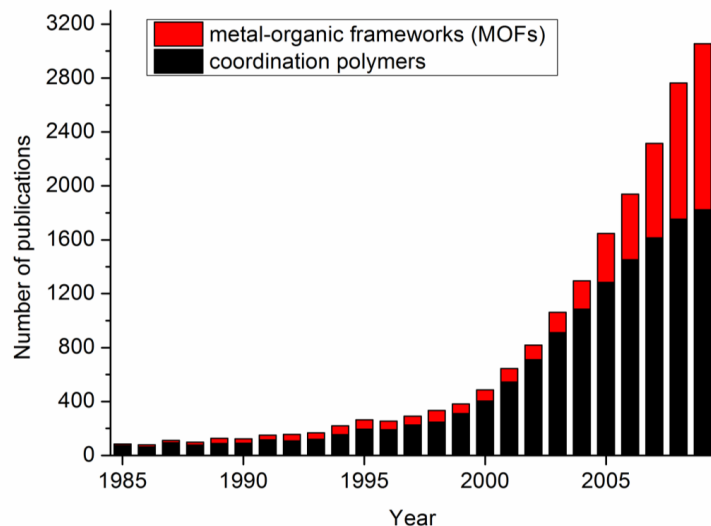
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Content

1. Definition and stability issue
2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look
5. Synthesis and analysis
6. Selected applications

Metal-organic frameworks (MOFs): A dynamic field

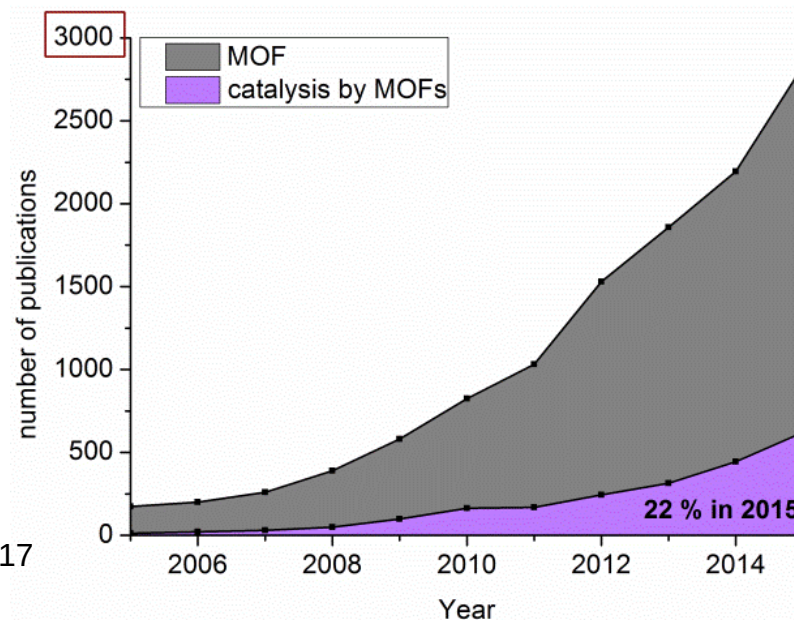


Scifinder: "metal organic framework", "MOF" or "coordination polymer"

C. Janiak, J. K. Vieth
New J. Chem. **2010**, 34, 2366-2388
DOI: 10.1039/C0NJ00275E

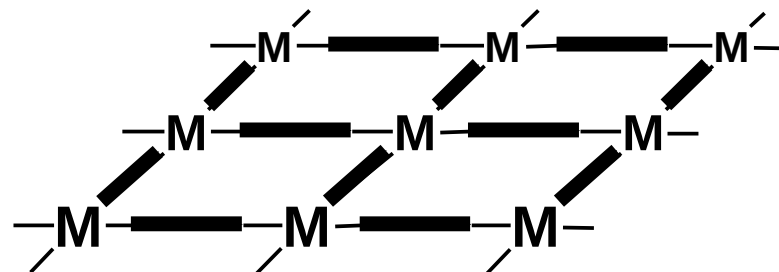
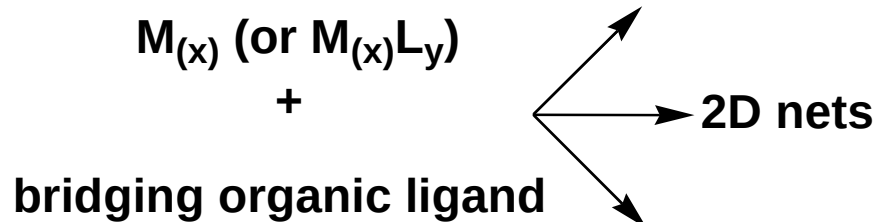
A. Herbst, C. Janiak,
CrystEngComm **2017**, 19, 4092-4117
DOI: 10.1039/C6CE01782G

Scifinder: "MOF", "catalysis by MOFs"

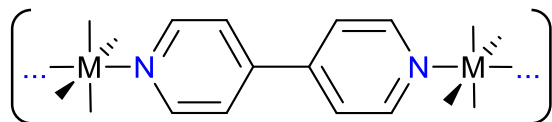
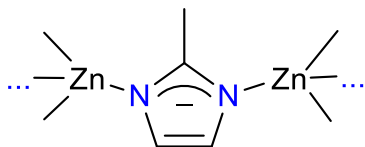
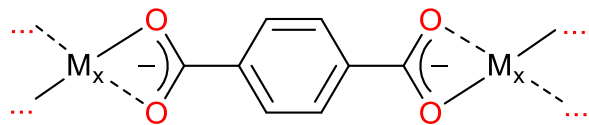


Coordination polymers / metal-organic frameworks (MOFs)

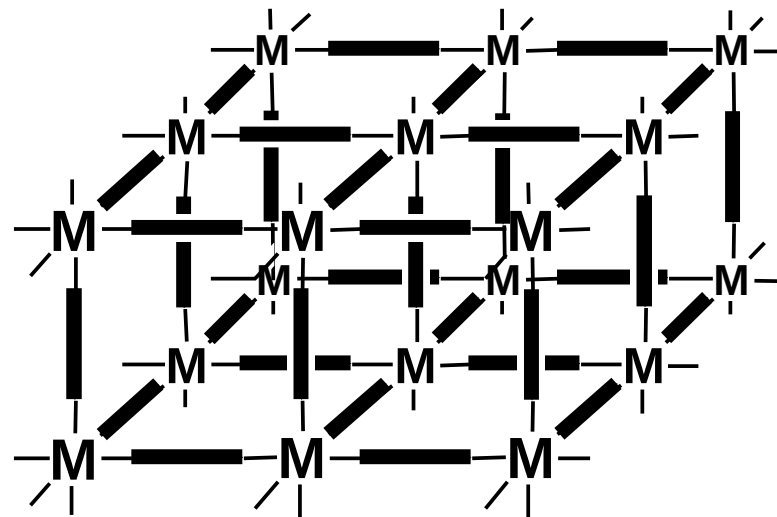
1D chains —M—M—M—M—



Examples:



3D frameworks



Metal-organic framework

- but no metal-carbon bond

until 1995:

metal-organic = organometallic = metal-carbon compounds

since introduction of "metal-organic framework", "MOF"

metal-organic = metal with organic ligand

- with O, N etc. donor atoms
- and coordinative M-O, M-N etc. bond

=> MOFs are coordination compounds

Coordination polymers / metal-organic frameworks (MOFs) – Definitions

Coordination polymer:

A **coordination compound** with repeating coordination entities extending **in 1, 2, or 3 dimensions**.

S. R. Batten, N. R. Champness, X.-M. Chen, J. G.-Martinez, S. Kitagawa, L. Öhrström, M. O'Keeffe, M. P. Suh, J. Reedijk, *CrystEngComm* **2012**, 14, 3001–3004;
Pure Appl. Chem. **2013**, 85, 1715-1724.

Coordination polymers / metal-organic frameworks (MOFs) – Definitions

Coordination polymer:

A **coordination compound** with repeating coordination entities extending **in 1, 2, or 3 dimensions**.

Coordination network:

A **coordination compound** extending, through repeating coordination entities, in 1 dimension, but with cross-links between two or more individual chains, loops, or spiro-links, or a coordination compound **extending** through repeating coordination entities **in 2 or 3 dimensions**.

S. R. Batten, N. R. Champness, X.-M. Chen, J. G.-Martinez, S. Kitagawa, L. Öhrström, M. O'Keeffe, M. P. Suh, J. Reedijk, *CrystEngComm* **2012**, 14, 3001–3004;
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Coordination polymers / metal-organic frameworks (MOFs)

– Definitions

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Metal-organic frameworks:

A metal–organic framework, abbreviated to MOF, is a coordination network with **organic ligands** containing **potential voids**.

not necessarily – **but most often**

- 3D
- crystalline
- permanent and proven porosity
- organic bridging ligand

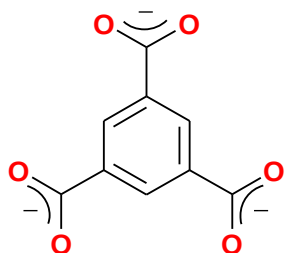
- MOFs have to be activated to achieve porosity
- crystalline nature allows for (single crystal) X-ray diffraction studies
- porosity determination by gas sorption (N₂, CO₂, H₂, Ar)

S. R. Batten, N. R. Champness, X.-M. Chen, J. G.-Martinez, S. Kitagawa, L. Öhrström, M. O’Keeffe, M. P. Suh, J. Reedijk, *CrystEngComm* **2012**, 14, 3001–3004;
Pure Appl. Chem. **2013**, 85, 1715-1724.

Metal-Organic Frameworks (MOFs)

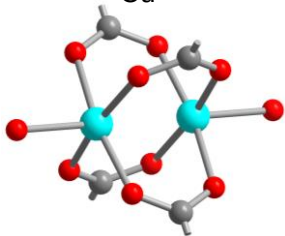
– 3 examples

prototypical linkers
(ligands)

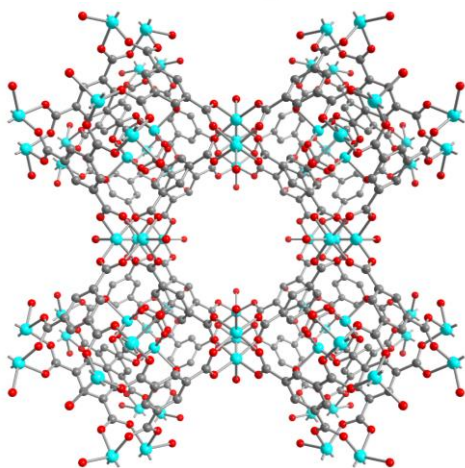


benzene-1,3,5-tricarboxylate,
trimesate, BTC
+
 Cu^{2+}

metal cluster =
secondary building
units (SBU)



3D networks



acronyms

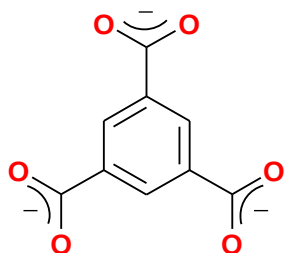
HKUST-1

- more details later

Metal-Organic Frameworks (MOFs)

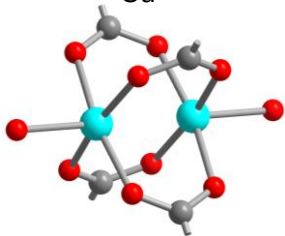
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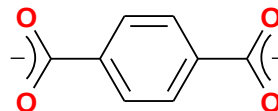


benzene-1,3,5-tricarboxylate,
trimesate, BTC
+ Cu^{2+}

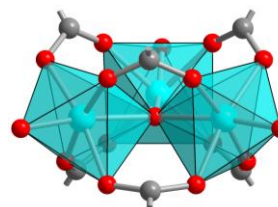
metal cluster =
secondary building
units (SBU)



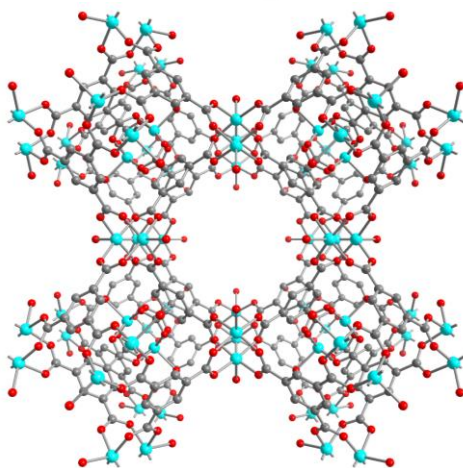
- more details later



benzene-1,4-dicarboxylate,
terephthalate, BDC
+ Cr^{3+}

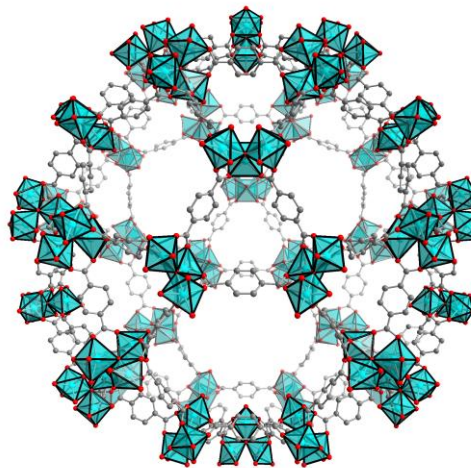


3D networks



acronyms

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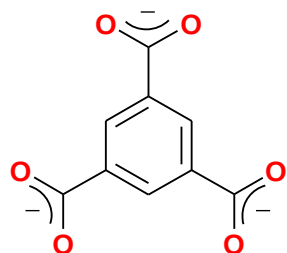


MIL-101

Metal-Organic Frameworks (MOFs)

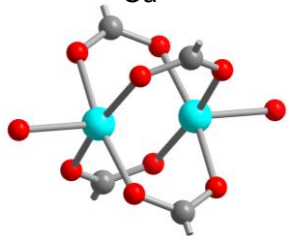
– 3 examples

prototypical linkers
(ligands)

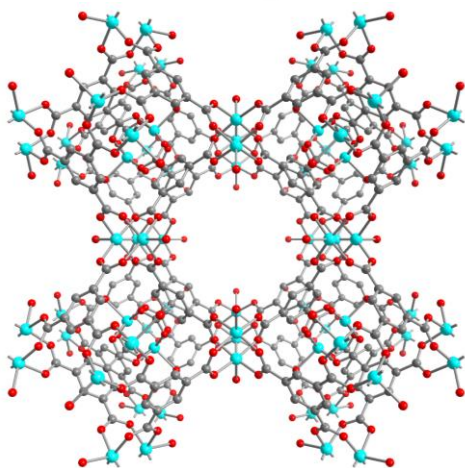


benzene-1,3,5-tricarboxylate,
trimesate, BTC

+
 Cu^{2+}



metal cluster =
secondary building
units (SBU)

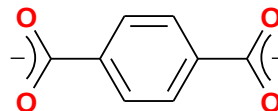


3D networks

acronyms

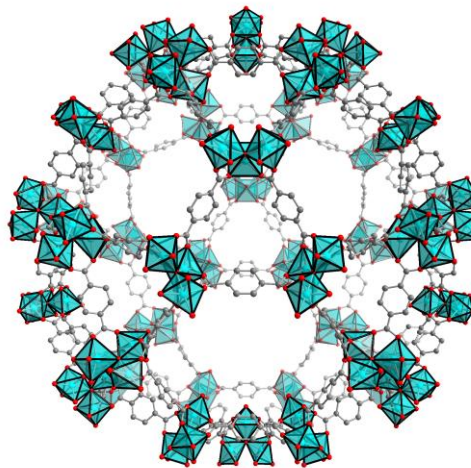
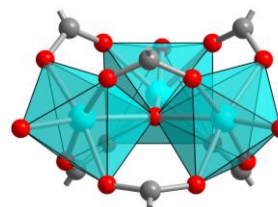
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- more details later

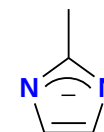


benzene-1,4-dicarboxylate,
terephthalate, BDC

+
 Cr^{3+}

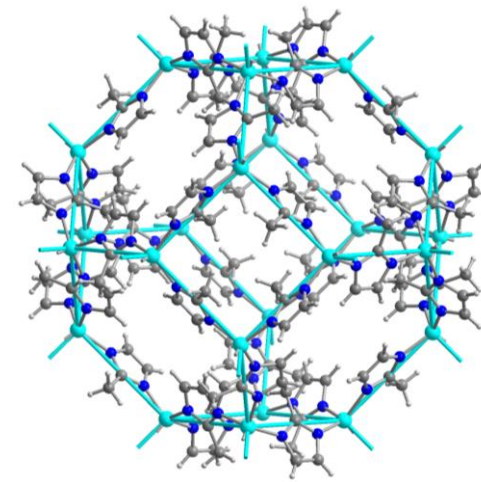
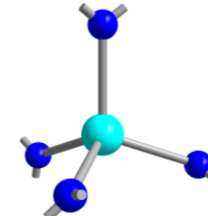


MIL-101



2-methyl-
imidazolate

+
 Zn^{2+}



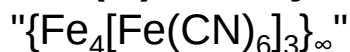
ZIF-8

Prussian Blue – coordination polymer and network

- but no MOF

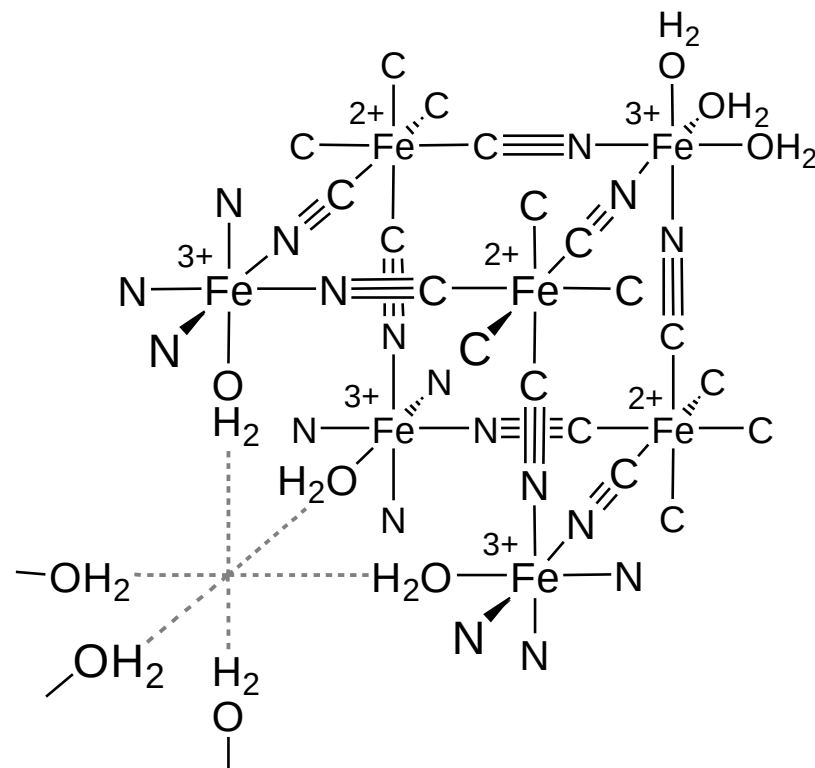
Prussian Blue

iron(III)-hexacyanidoferrate(II)



- three-dimensional (3D)
- mixed-valent
- coordination polymer;
- cyanide, CN^- bridges between octahedrally coordinated Fe-atoms;
- Fe^{2+} is C-coordinated,
- Fe^{3+} is N-coordinated by the bridging ligand;
- porous (Tl^+ or Cs^+ can be incorporated by exchange with remaining K^+ ions)

- but no organic ligands

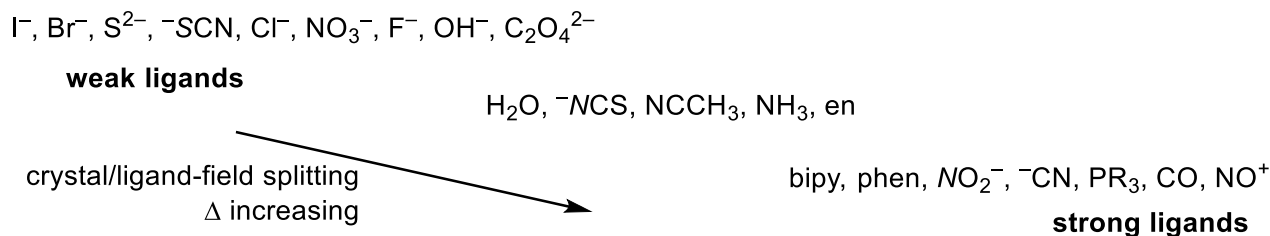


colloidal Prussian Blue

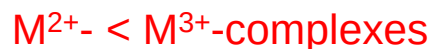
- as medical therapeutic, oral doses of up to 20 g/day
- antidote against thallium poisoning and
- for decorporation (elimination from the body) or preventing resorption of radioactive cesium (^{137}Cs)
- trade names: Antidotum Thallii Heyl®, Radiogardase®-Cs

Important: MOFs are coordination compounds

- for the M-L coordinative bond coordination chemistry principles apply
- spectrochemical series



- stability trends:



Important: MOFs are coordination compounds

HSAB concept

Metal cations – acids	Ligands – bases
hard acids	hard bases
- highly charged e.g.	e.g.
Al^{3+} , Cr^{3+} , Fe^{3+} , Co^{3+} Ti^{4+} , Zr^{4+} , Hf^{4+}	F^- O-ligands amine ligands
soft acids	soft bases
Pd^{2+} , Pt^{2+} Cu^+ , Ag^+ Cd^{2+}	I^- S-ligands P-ligands C-ligands

Important: MOFs are coordination compounds

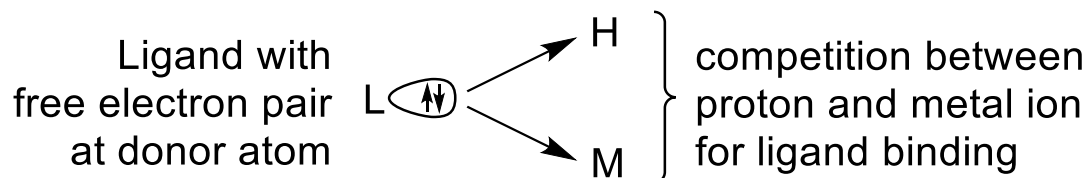
(thermodynamics)

- formation/dissociation equilibrium:

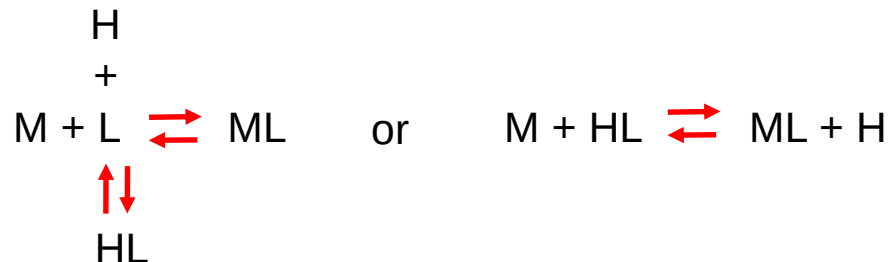


H_2O is a good ligand!

- ligands are bases:



=> pH dependency:



(charges are not shown
for simplification)

Important: MOFs are coordination compounds

(kinetics)

- substitution **labile and inert** metal-ligand bonds

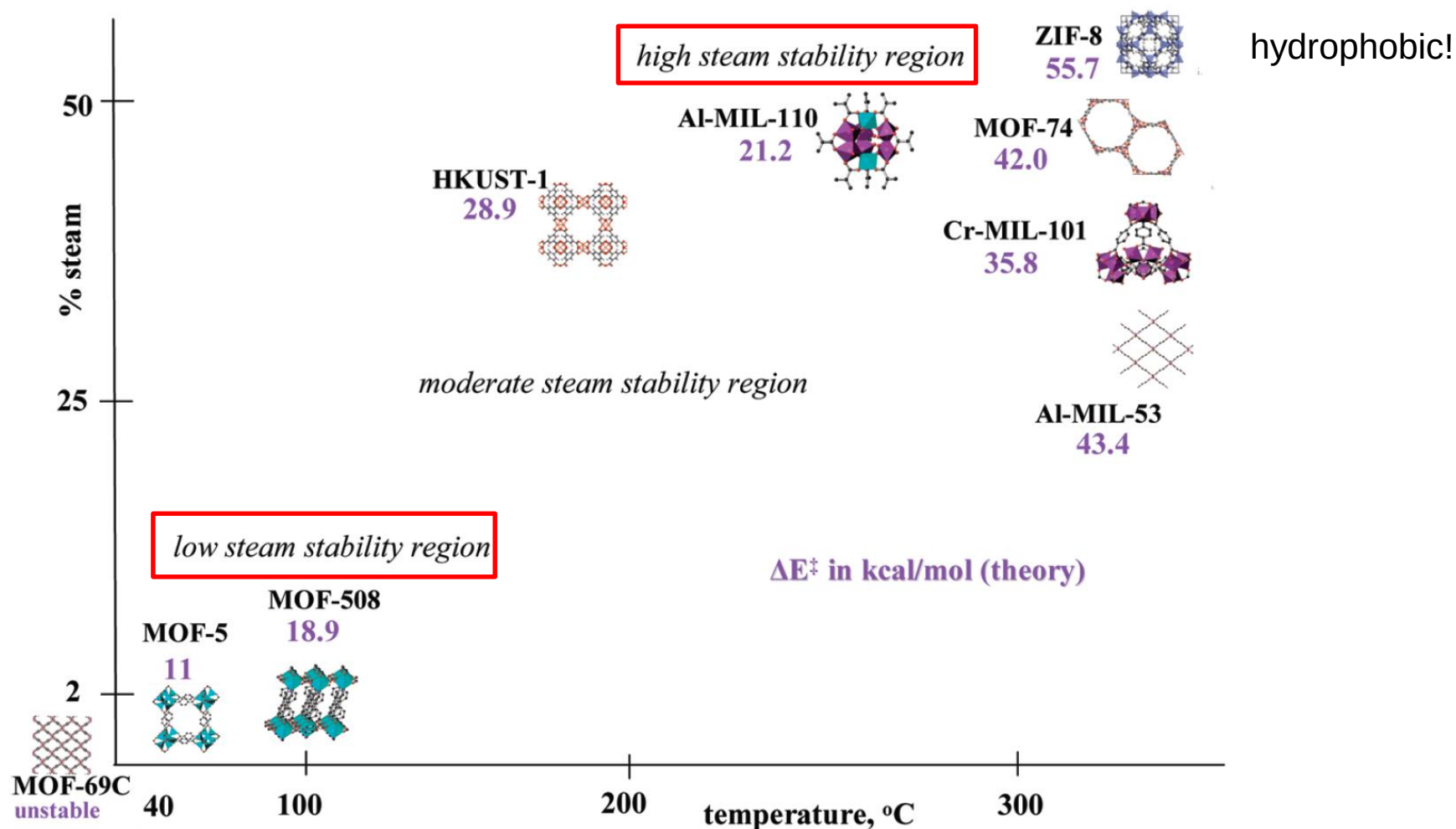
inert **metal ions**: Cr^{3+} , Co^{3+} with d^3 , d^6 (low-spin) configuration

labile metal ions: most others, e.g. Zn^{2+}

- **more covalent M-L bonds are more inert** than more ionic bonds
e.g. M-N more inert compared to M-O more labile

Water stability of MOFs

- an inherent problem (as water is almost everywhere)



J. J. Low, A. I. Benin, P. Jakubczak, J. F. Abrahamian, S. A. Faheem, R. R. Willis,
J. Am. Chem. Soc. **2009**, 131, 15834.

Water stability of MOFs

- an inherent problem (as water is almost everywhere)

	3 days												2 months													
	pH=0		pH=4		pH=12		H ₂ O ₂		H ₂ O		air		pH=0		pH=4		pH=12		H ₂ O ₂		H ₂ O		air		Steam test	
	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂
MIL-101 (Cr)																										
NH ₂ -MIL-101 (Al)																										
MIL-53 (Al)																										
NH ₂ -MIL-53		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP
UiO-66																										
NH ₂ -UiO-66																										
UiO-67																										
CuBTC																										
ZIF-8																										

(indicated
MOFs are
described later)

Powder X-ray diffractogram
and surface area, S_{BET}

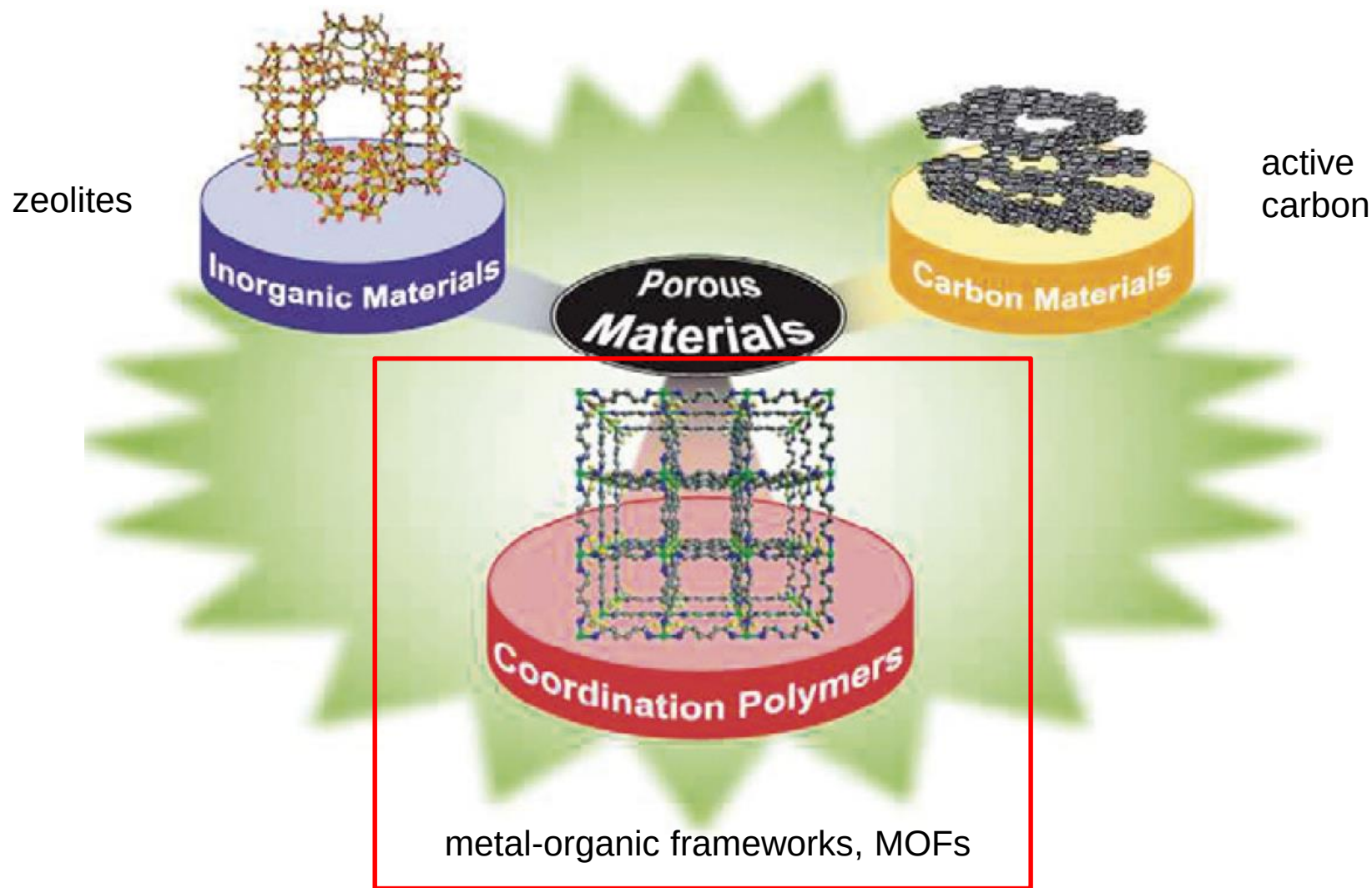
 completely
 partially
destroyed

K. Leus, T. Bogaerts, J. De Decker, H. Depauw,
K. Hendrickx, H. Vrielinck, V. Van Speybroeck,
P. Van Der Voort,
Micropor. Mesopor. Mater. **2016**, 226, 110-116

Content

1. Definition
2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look
5. Synthesis
6. Selected applications

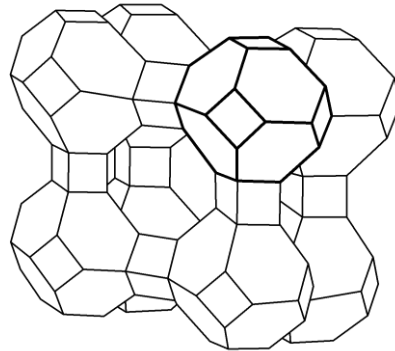
Metal-Organic Frameworks (MOFs) as a new porous material class



Porous materials

inorganic:

zeolites,
silica gel,
mesoporous silica



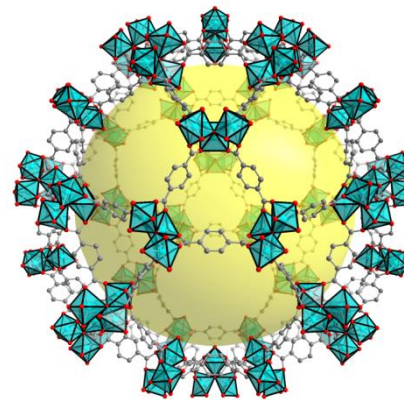
hygroscopic

applications as
desiccant

inorganic-organic hybrids:

Metal-Organic Frameworks, MOFs

- defined, identical pores
- designable properties



Nano-
pores

3.4 nm

organic :

active carbon

new: porous-organic polymers

e.g. Covalent-Organic Frameworks, COFs



Porous materials: specific surface area (BET) per gram of material

inorganic:

zeolites,
silica gel,
mesoporous silica ~1000 m²/g

inorganic-organic hybrids:

Metal-Organic Frameworks, MOFs

~2000-4000 m²/g

organic :

active carbon
~1000 m²/g



What is special of MOFs? among porous materials

- higher surface area
- crystallinity => exact structure can be determined
 - pore system very well defined
 - narrow range of pore distribution
- organic linkers give great range of design options
- wide variation of metal atoms and organic linkers to tune properties
 - hydrophilicity – hydrophobicity
 - pore sizes
 - incorporation of chirality
 - electronic properties

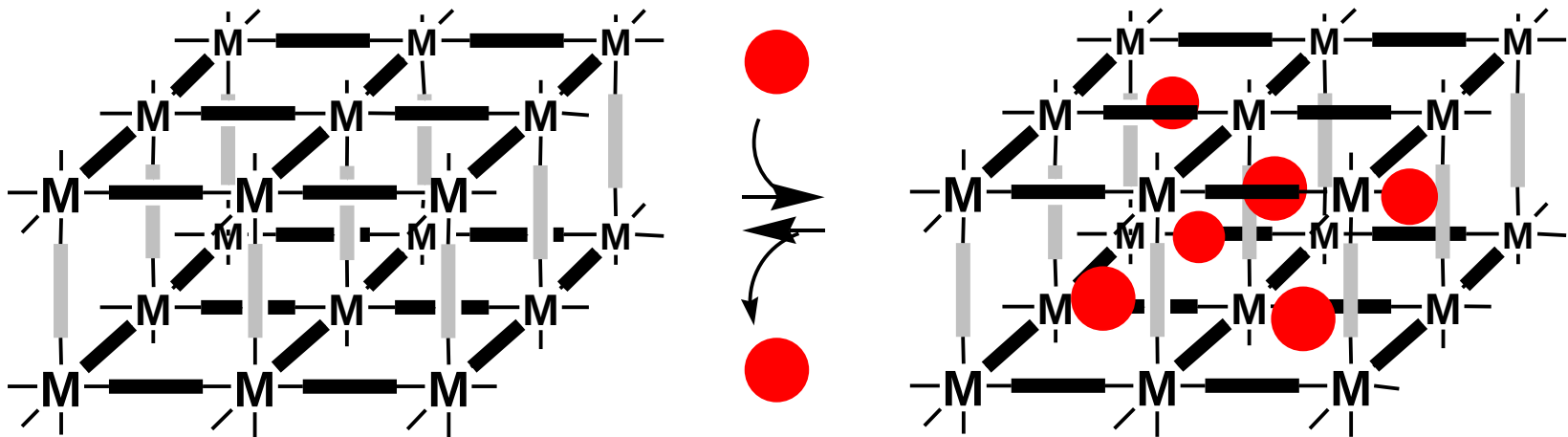
silica gel, active carbon,
mesoporous silica
- amorphous
- wider range of pores

zeolites, silica gel,
mesoporous silica,
active carbon
- very limited range of
building blocks

Aspect of porosity

- reversible guest exchange / storage / transformation

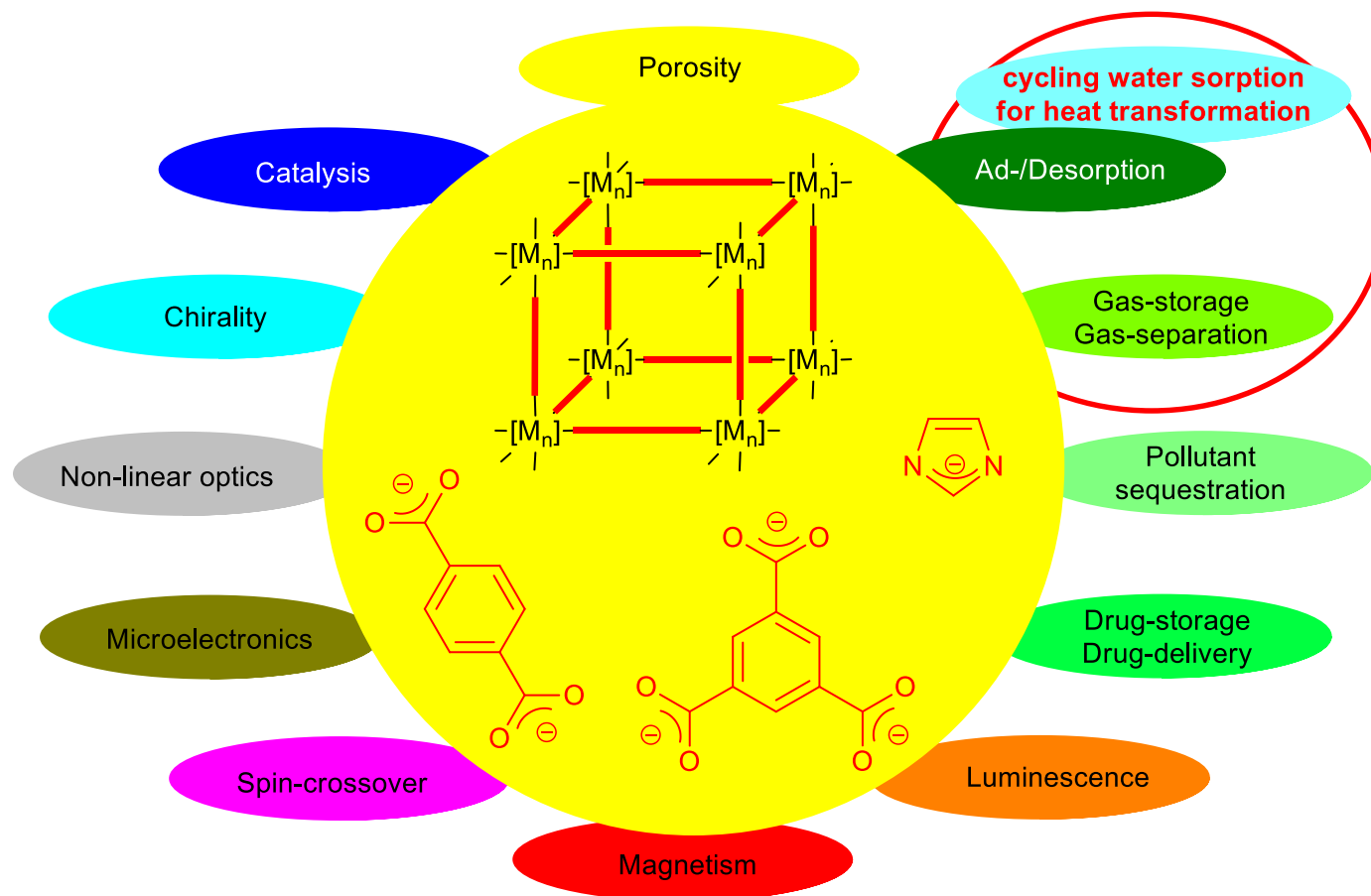
MOFs = "organic zeolite analogs"



Applications of zeolites, silica gel,
mesoporous silica, active carbon:

- ad/desorption
- ion exchange (e.g. water softening in detergents)
- separation (chromatography)
- drying agent ("molecular sieve")
- selective catalysis (petrochemical refining, cracking)
- etc.

Metal-Organic Frameworks (MOFs): Application-oriented properties



Reviews: *New J. Chem.* **2010**, 34, 2366 DOI: 10.1039/C0NJ00275E.
Dalton Trans. **2003**, 2781 DOI: 10.1039/b305705b.

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1. Definition
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MOF history

- **starting ~1990**
- an excellent example of interdisciplinary research
- **first molecule-based** field of **crystal engineering / coordination chemistry**
- **later zeolite chemists**, solid-state chemists, material scientists
- coordination polymers preceded MOFs:
 - term **coordination polymers** coined and area first reviewed in **1964**.

Coordination polymers – first(?) notion

CHAPTER 1

Coordination Polymers

JOHN C. BAILAR, JR.

University of Illinois, Urbana, Illinois

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from: Preparative Inorganic Reactions, ed. W. L. Jolly, Interscience, New York, 1964, vol. 1, pp. 1-25.

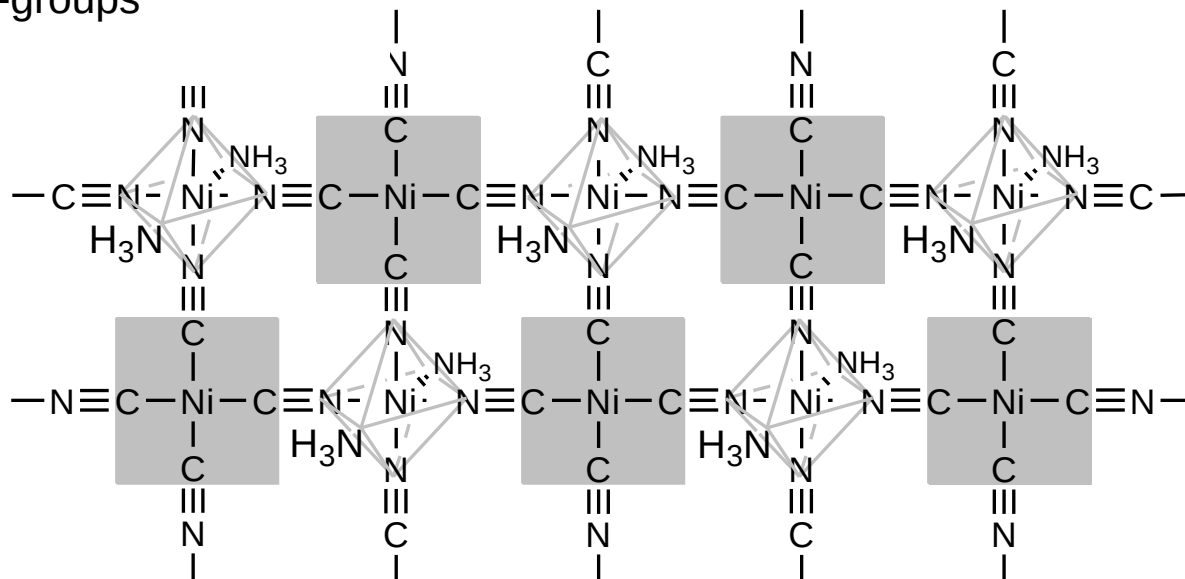
MOF history

- term **coordination polymers** coined and area first reviewed in 1964.
 - **interest in porous coordination polymers came much later,**
 - although some coordination complexes such as
 - Prussian blue compounds (see above) and
 - Hofmann clathrates
- had been known to show reversible sorption properties.

Hofmann clathrates

2D cyanido-metal framework, constructed from

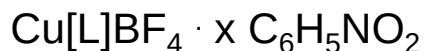
- alternating square-planar $\text{Ni}^{\text{II}}(\text{CN})_4$ -units and
- octahedral $\text{Ni}^{\text{II}}(\text{NC})_4(\text{NH}_3)_2$ -groups



- guest molecules are not shown
- guests can be organic molecules, especially aromatics
- reversible guest removal can be possible
- instead of Ni in the octahedral position also Cd can be used.
- when bridging amine ligands, e.g., 4,4'-bipyridine are used in place of NH_3 -groups then the 2D layers are vertically interconnected to a 3D network

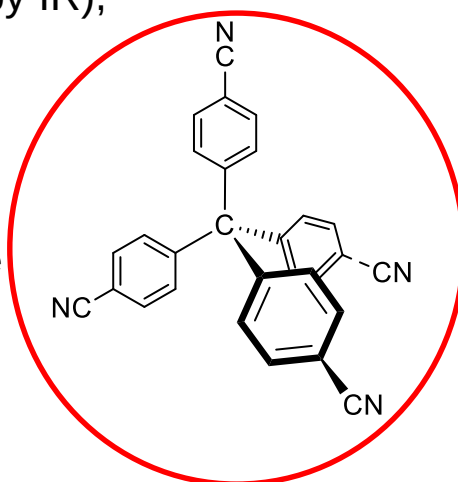
MOF history

interest in **porous coordination polymers** and MOFs started only **around 1990**.
-work by **Hoskins and Robson** (Univ. Melbourne) set the basis for the future of MOFs.



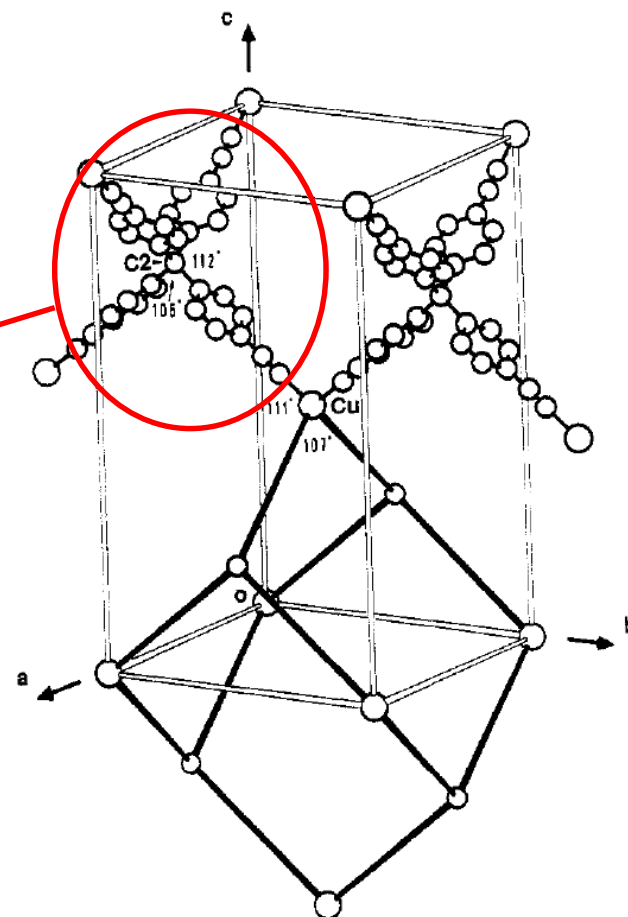
- anion exchange with PF_6^- (by IR),
- diamond-related framework
- large adamantane-like cavities of $\sim 1500 \text{ \AA}^3$

tetracyanotetraphenylmethane



Hoskins and Robson envisioned

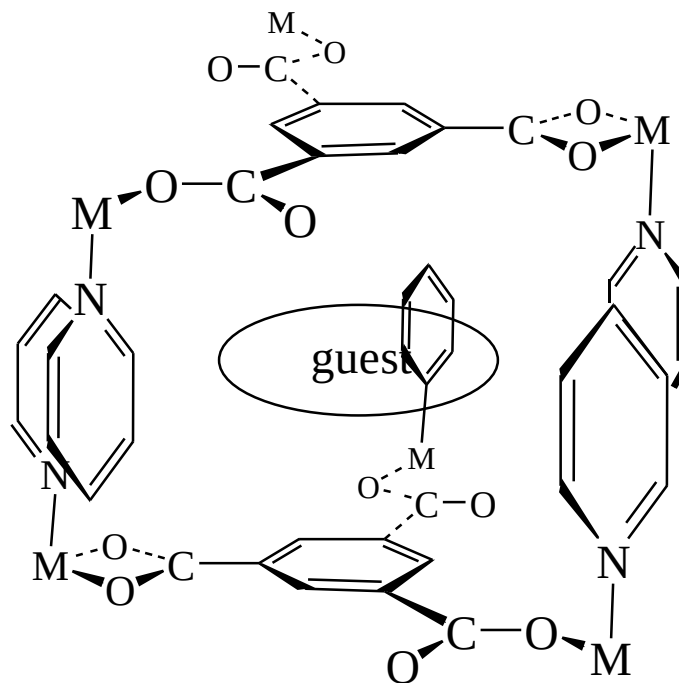
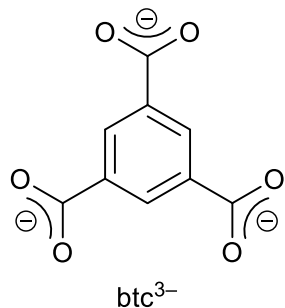
- large range of **crystalline, microporous, stable solids**,
- possibly using structure-directing agents,
- with **ion-exchange, gas sorption, or catalytic properties**,
- postsynthetic modifications**



MOF history

The term **MOF** was popularized by Yaghi et al. around **1995**

for a layered Co-trimesate that showed reversible sorption properties.



MOF history

Framework integrity in the absence of guest molecules?

-layer structure, with the pyridine ligands functioning as spacer units between the layers.

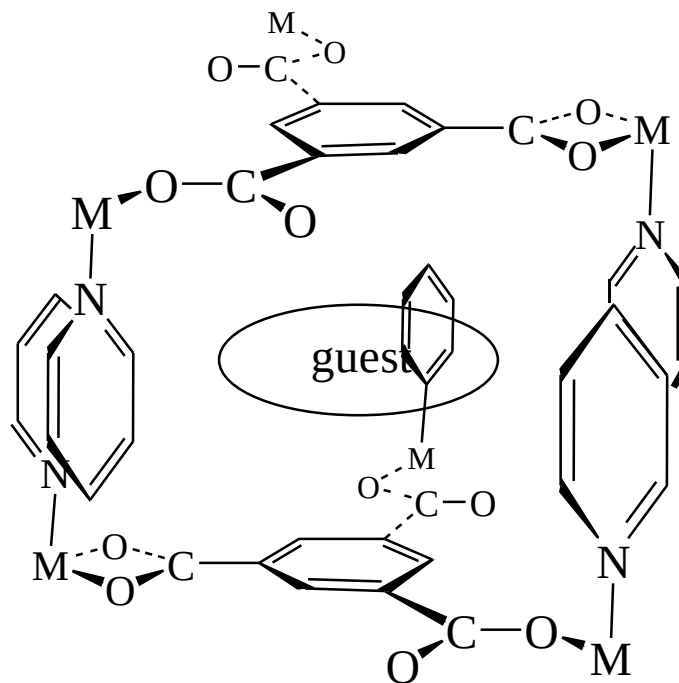
-7x10 Å rectangular channels,

-aromatic guest can reversibly be removed.

- single crystals retain their morphology and crystallinity upon loss of the guests according to optical microscopy.

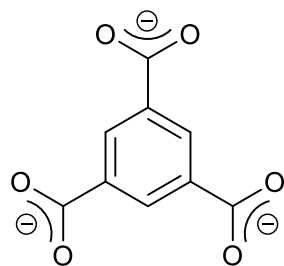
- from mixtures of components only the aromatic molecules such as benzene, nitrobenzene, cyano- or chlorobenzene are selectively absorbed (by IR).

-crystal lattice is thermally stable up to 350 °C.

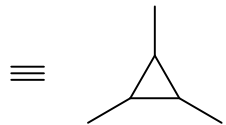


O. M. Yaghi, G. Li and H. Li, *Nature*, 1995, **378**, 703-706.

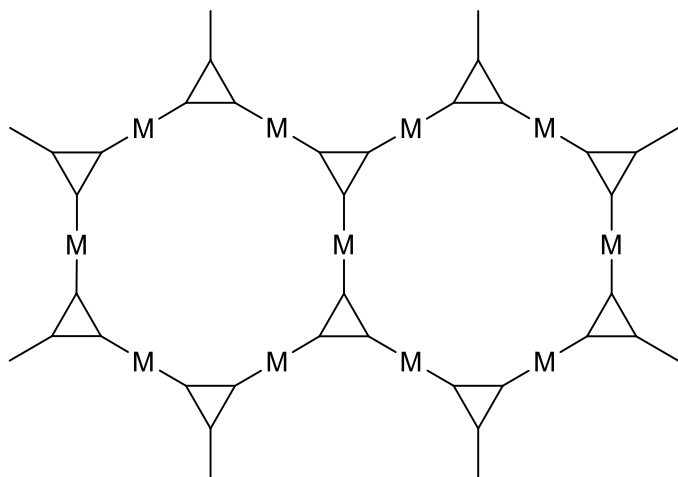
MOF history



1, btc^{3-}



$\text{M} = \text{Co}^{2+}, \text{Ni}^{2+} \text{ and } \text{Zn}^{2+}$



2

Rigorous proof of framework integrity in the absence of guest molecules still needed to be developed.

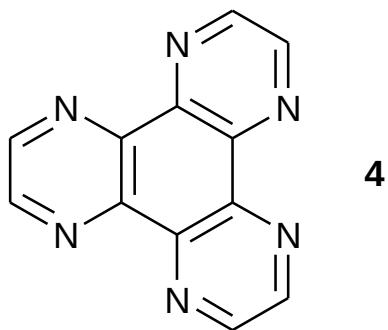
hydrated solid of formula $\text{M}_3(\text{btc})_2 \cdot 12\text{H}_2\text{O}$

-2D network structure with 1D channels

- reversibly and repeatedly binds water,
- inclusion of ammonia was demonstrated,
- larger molecules and those without a reactive lone pair did not enter

-retention of the 2D metal-ligand assembly with 1D channels (**2**) *was proposed* for the porous solid

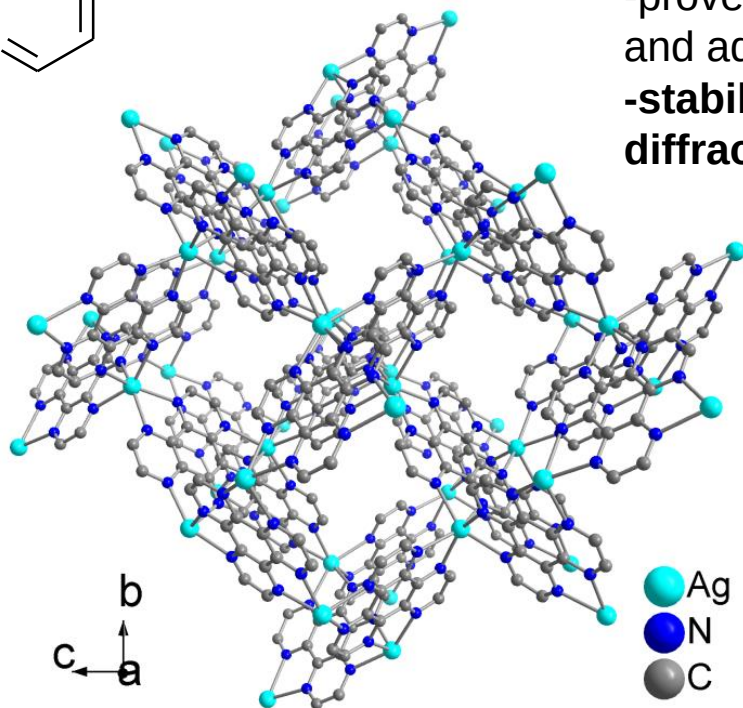
MOF history



Rigorous proof of framework integrity!

three-dimensional 10,3-a network from Ag(I) and ligand 1,4,5,8,9,12-hexaazatriphenylene (hat, **4**)
-proved stable over two nitromethane guest de- and adsorption cycles

-stability evidenced by single-crystal X-ray diffraction over four generations.

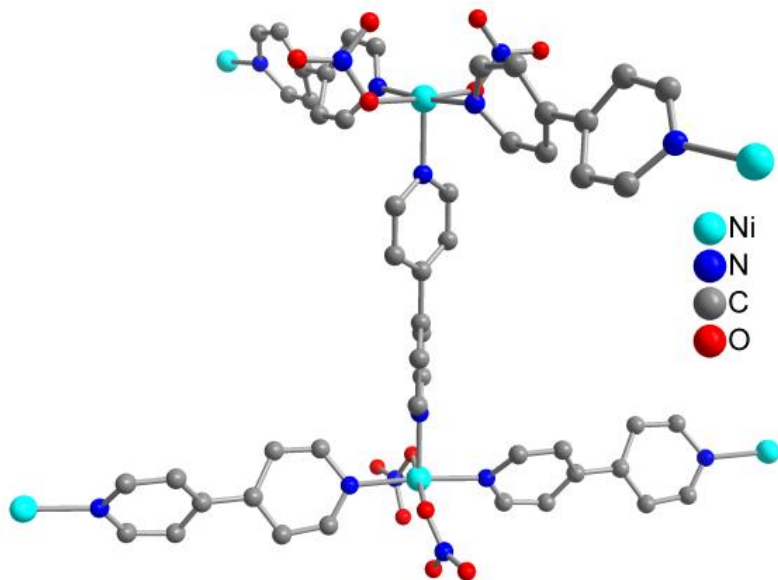


B. F. Abrahams, P. A. Jackson, R. R. Robson, *Angew. Chem. Int. Ed. Engl.*, **1998**, 37, 2656-2659.

MOF history

Rigorous proof of framework integrity!

Kepert and Rosseinsky proved the rigidity of the two-dimensional (2D) coordination polymer network of $[\text{Ni}_2(4,4'\text{-bipy})_3(\text{NO}_3)_4]$ (bipy = bipyridine) quantitatively by single-crystal X-ray structures of both ethanol-loaded and ethanol-free forms.



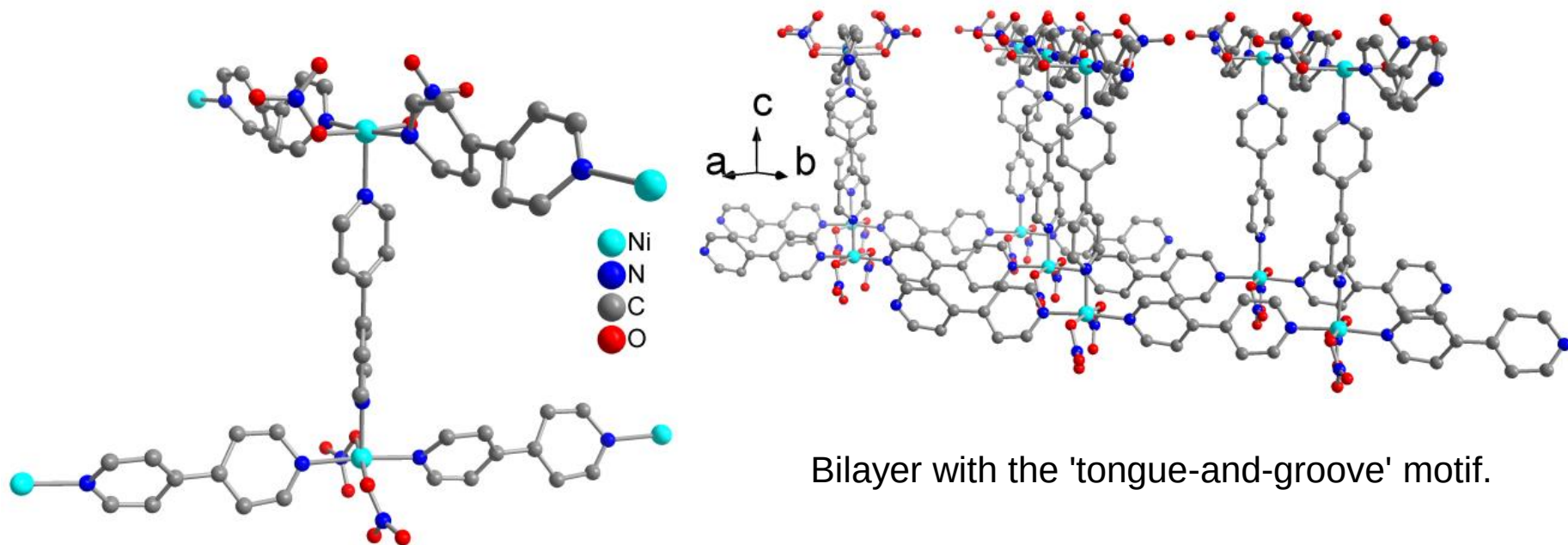
Extended building block of the 2D network $[\text{Ni}_2(4,4'\text{-bipy})_3(\text{NO}_3)_4]$ formed by T-shaped coordination of 4,4'-bipyridine to Ni^{2+} with nitrate ligands.

C. J. Kepert and M. J. Rosseinsky, *Chem. Commun.*, **1999**, 375-376.

MOF history

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Kepert and Rosseinsky proved the rigidity of the two-dimensional (2D) coordination polymer network of $[\text{Ni}_2(4,4'\text{-bipy})_3(\text{NO}_3)_4]$ (bipy = bipyridine) quantitatively by single-crystal X-ray structures of both ethanol-loaded and ethanol-free forms.



Bilayer with the 'tongue-and-groove' motif.

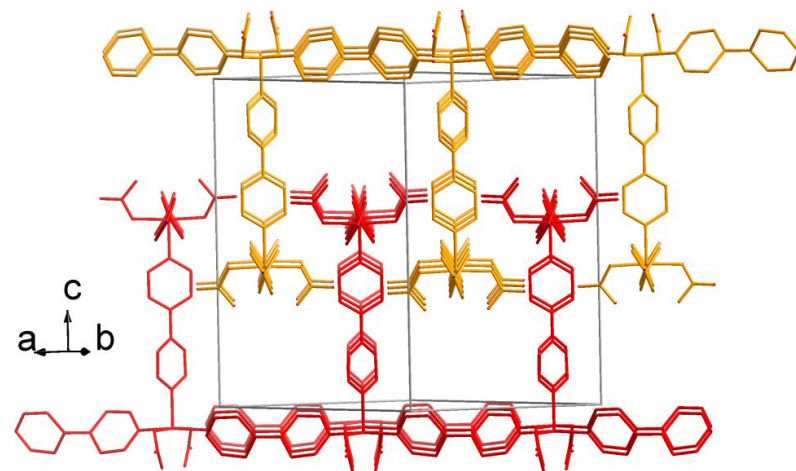
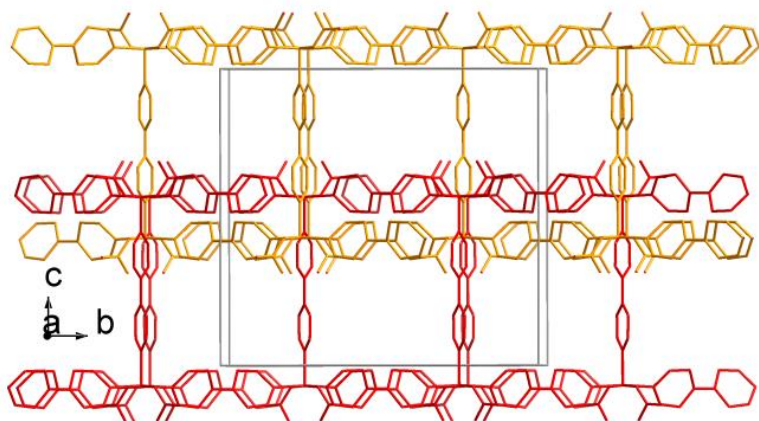
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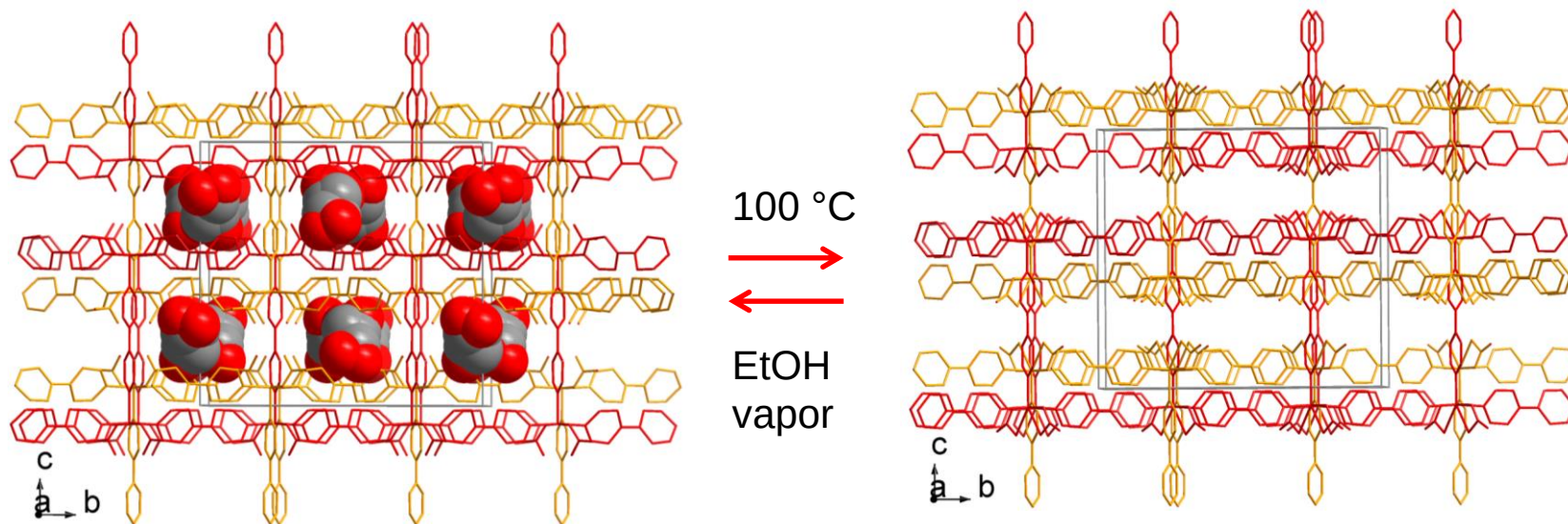
Interdigitation of two adjacent bilayers which are differentiated by color (in wire mode and two different viewing directions).

C. J. Keper and M. J. Rosseinsky, *Chem. Commun.*, **1999**, 375-376.

MOF history

Rigorous proof of framework integrity!

Keper and Rosseinsky proved the rigidity of the two-dimensional (2D) coordination polymer network of $[\text{Ni}_2(4,4'\text{-bipy})_3(\text{NO}_3)_4]$ (bipy = bipyridine) quantitatively by single-crystal X-ray structures of both ethanol-loaded and ethanol-free forms.

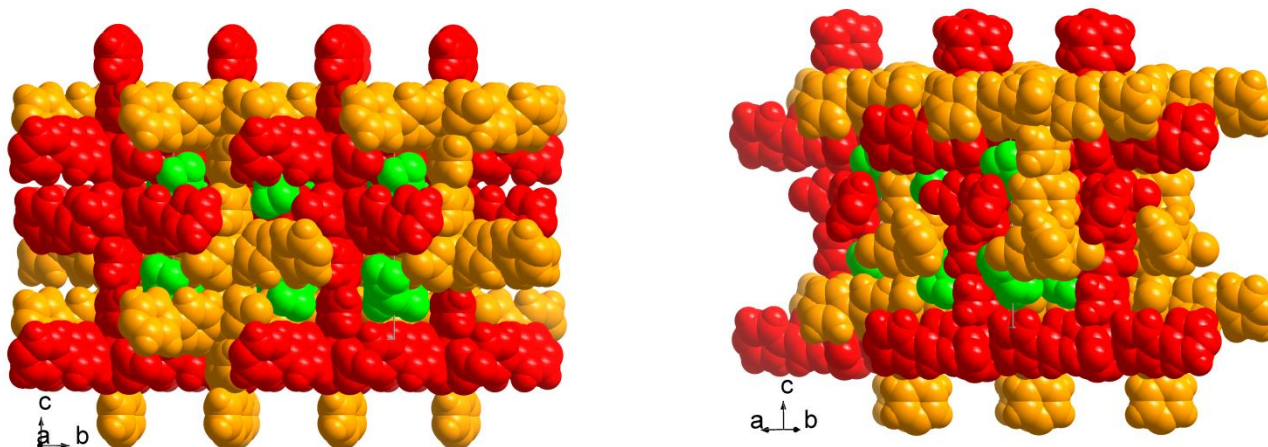


Ethanol-solvated and -emptied form. Channels with cavities run along the crystallographic *a* direction and occupy ~20% of the unit cell volume in both forms. Disordered ethanol molecules are shown in space-filling mode.

MOF history

Rigorous proof of framework integrity!

Kepert and Rosseinsky proved the rigidity of the two-dimensional (2D) coordination polymer network of $[\text{Ni}_2(4,4'\text{-bipy})_3(\text{NO}_3)_4]$ (bipy = bipyridine) quantitatively by single-crystal X-ray structures of both ethanol-loaded and ethanol-free forms.



A closer examination of the structure in space-filling mode reveals that the $[\text{Ni}_2(4,4'\text{-bipy})_3(\text{NO}_3)_4]$ network has only very small channels with dimension of ca. $1.1 \text{ \AA} \times 1.1 \text{ \AA}$. Therefore, the guest molecules are rather located in cavities ("zero-dimensional closed space"). => **Framework flexibility for ad/desorption necessary.**

Network of the ethanol-solvated structure of $[\text{Ni}_2(4,4'\text{-bipy})_3(\text{NO}_3)_4]$ in two viewing directions; the ethanol solvent molecules are depicted in green color.

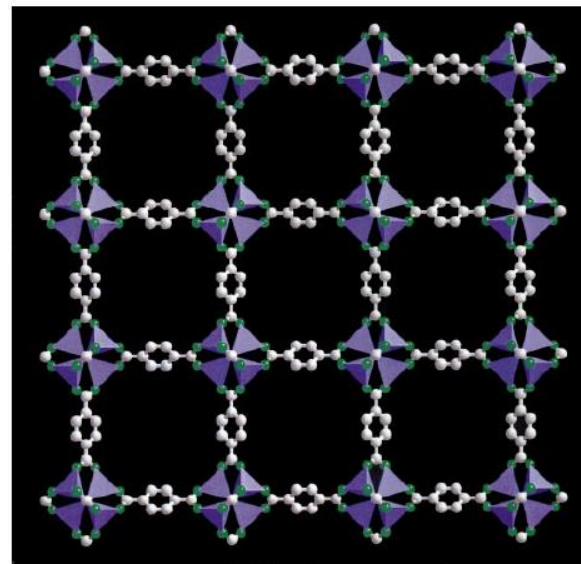
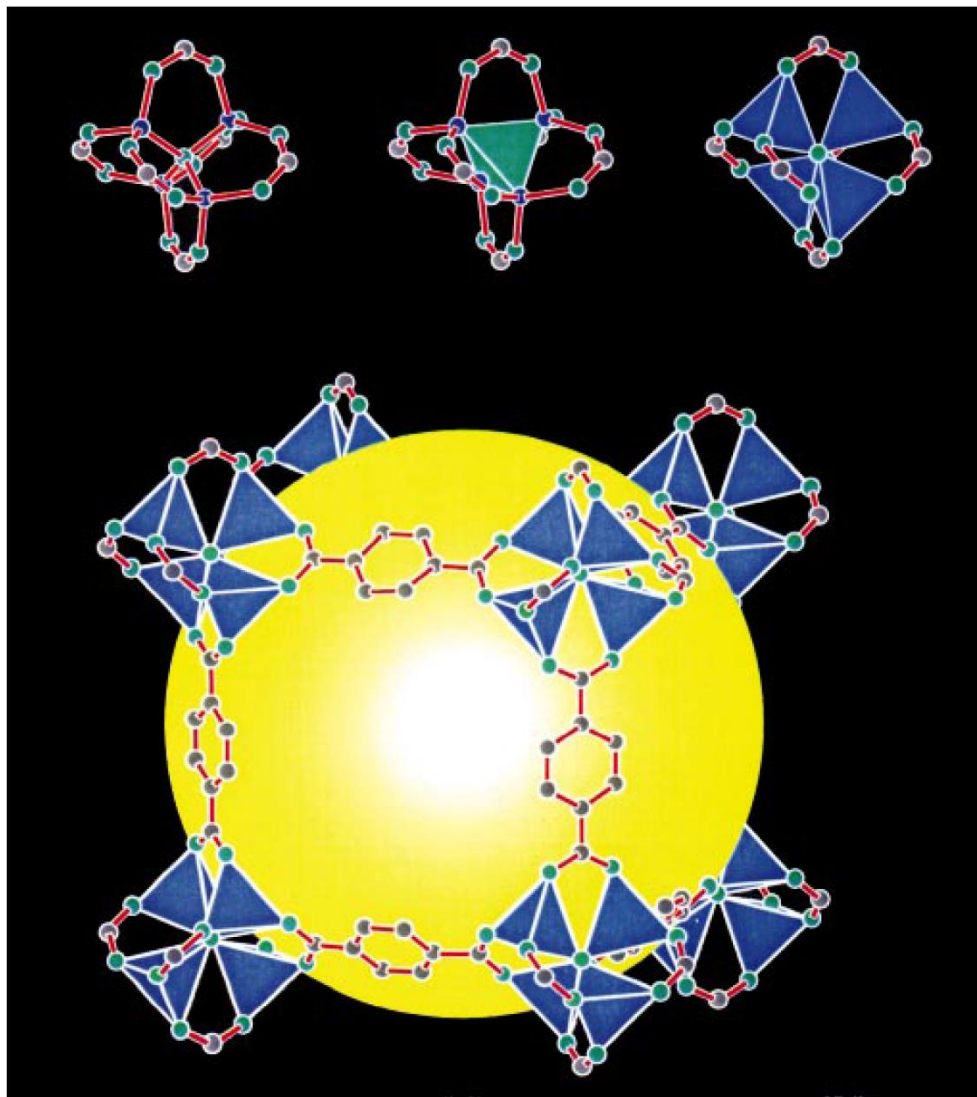
C. J. Kepert and M. J. Rosseinsky, *Chem. Commun.*, 1999, 375-376.

MOF history

In **1997**, a 3D MOF was reported by Kitagawa et al. that exhibited gas sorption properties at room temperature.

In **1999**, the syntheses of the prototypical **MOF-5** and **HKUST-1** were reported that in the following years and up to now are among the most studied MOFs.

MOF example:
 $3\text{D}[\text{Zn}_4(\mu_4\text{-O})(\text{bdc})_3]$ (also called MOF-5 or IRMOF-1)



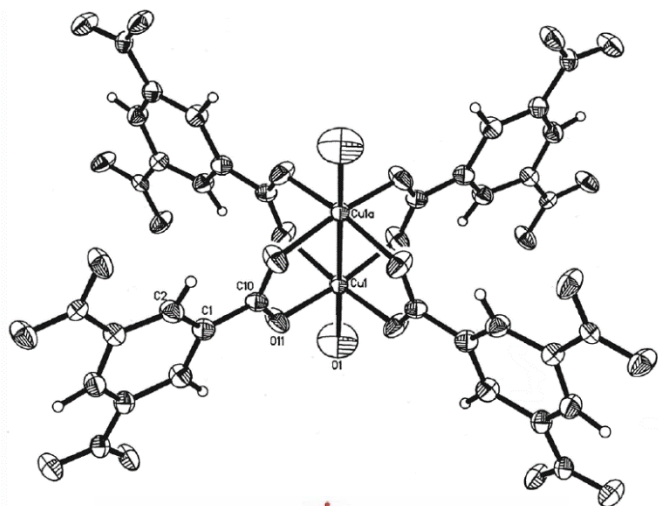
Langmuir surface area
 $\sim 2900 \text{ m}^2/\text{g}$

H. Li, M. Eddaoudi, M. O'Keeffe, O. M. Yaghi, *Nature* **1999**, 402, 276-279.

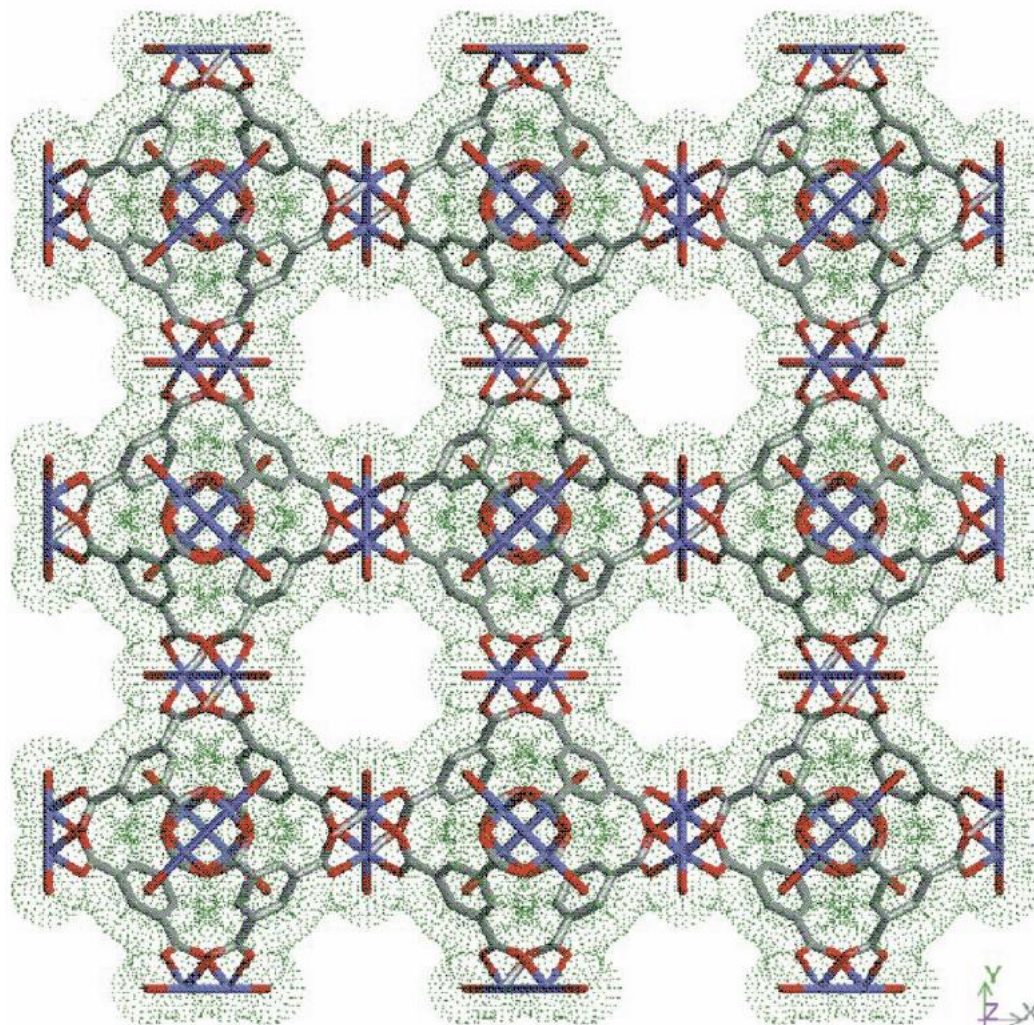
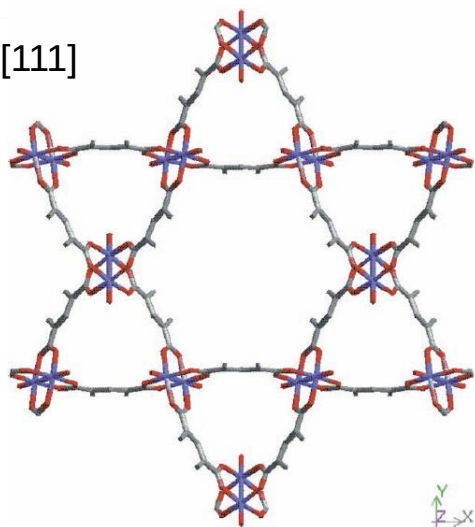
MOF example:

HKUST = Hongkong University of Science and Technology

along [100]



along [111]



S. S.-Y. Chui, S. M.-F. Lo, J. P. H. Charmant, A. G. Orpen, I. D. Williams, *Science* **1999**, 283, 1148-1150.

MOF history

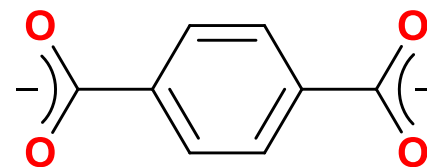
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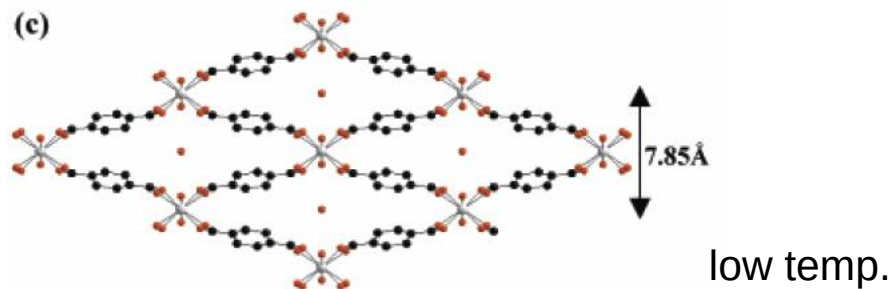
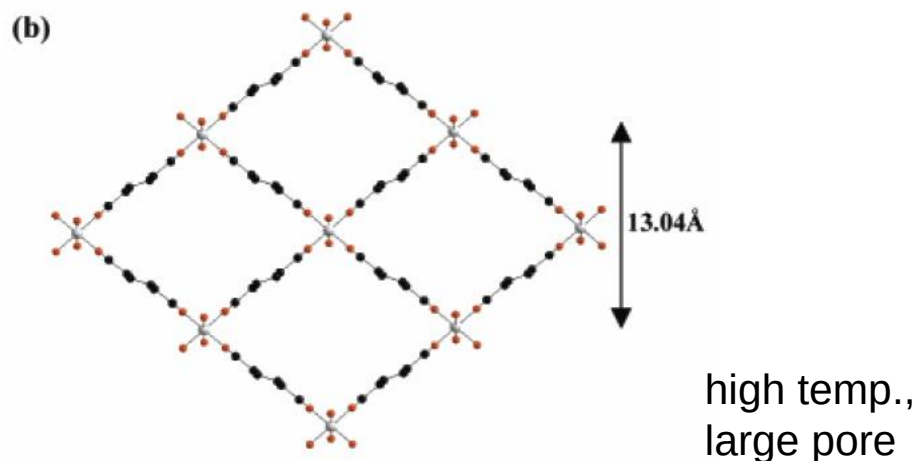
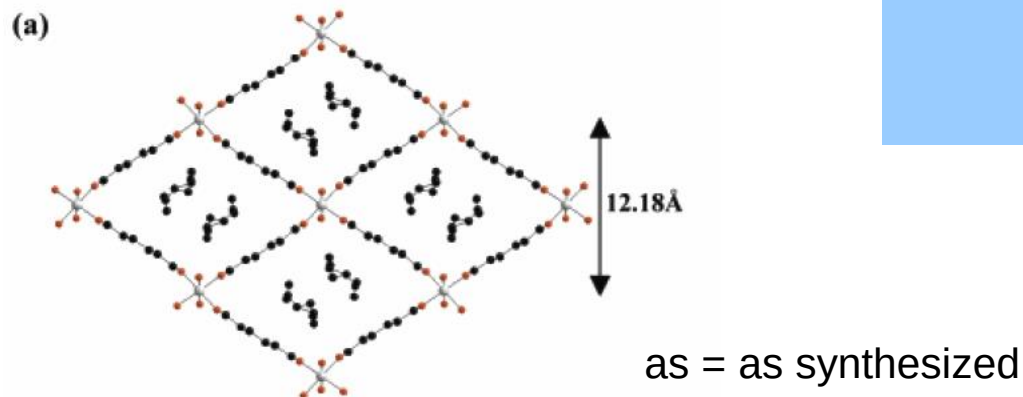
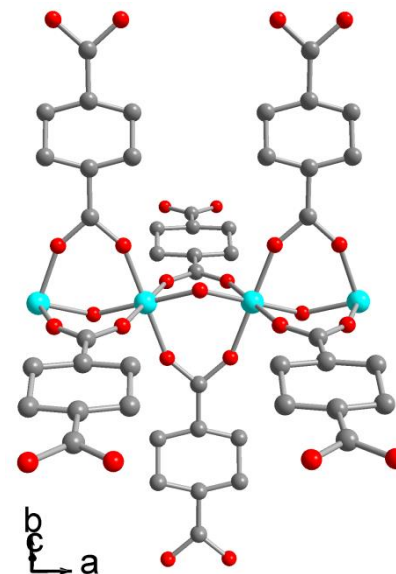
Starting in **2002**, Férey et al. reported porous MOFs, which he called **MILs**, like MIL-53 and MIL-88

MILs

3D-[Cr(μ_4 -bdc)(μ -OH)]



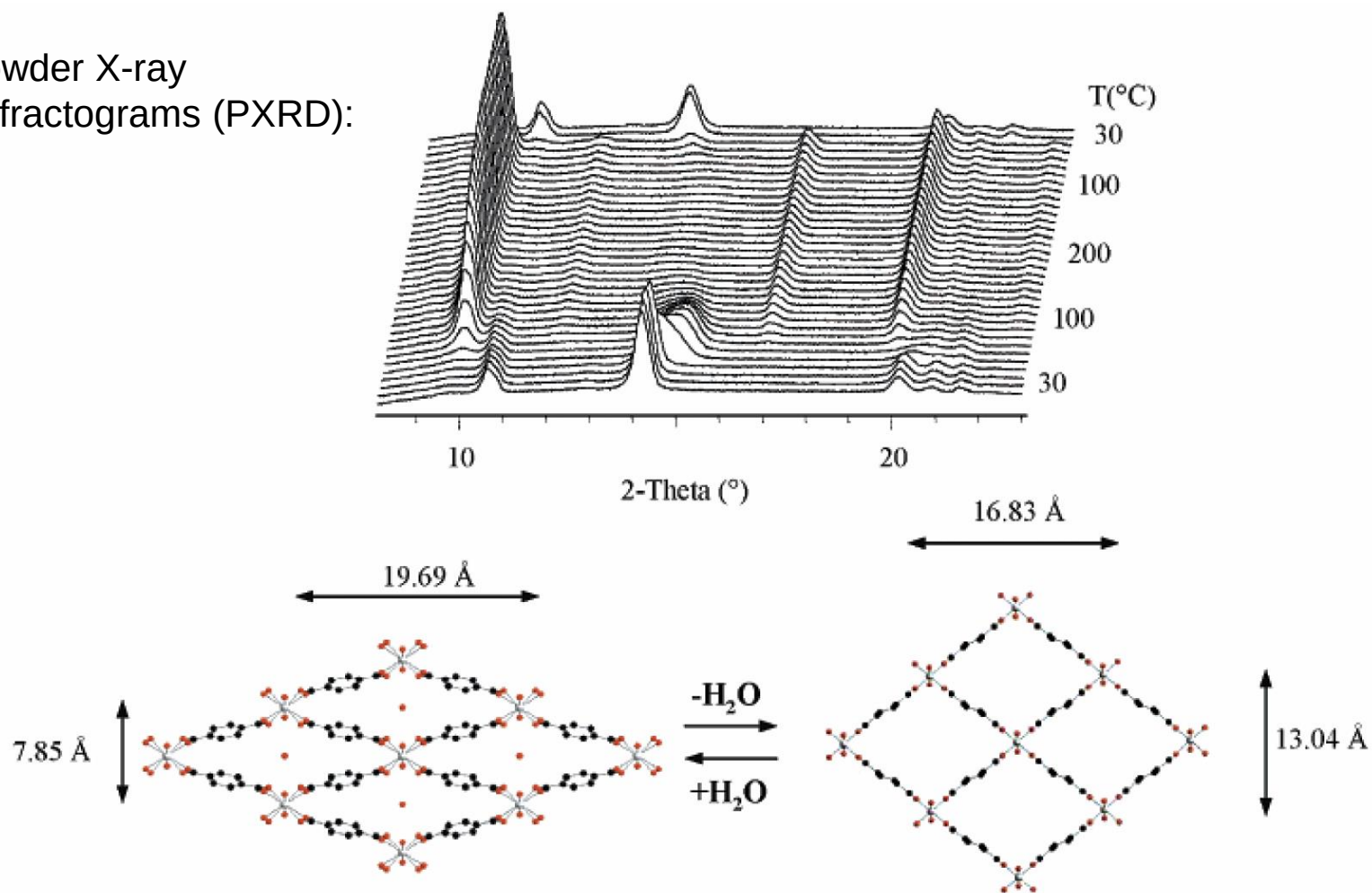
benzene-1,4-dicarboxylate,
terephthalate, bdc



MIL-53: Prototypical **breathing** MOF

3D-[Cr(μ_4 -bdc)(μ -OH)]

powder X-ray
diffractograms (PXRD):



C. Serre, F. Millange, C. Thouvenot, M. Nogues, G. Marsolier, D. Louer, G. Férey, *J. Am. Chem. Soc.* **2002**, *124*, 13519-13526.

MOF history

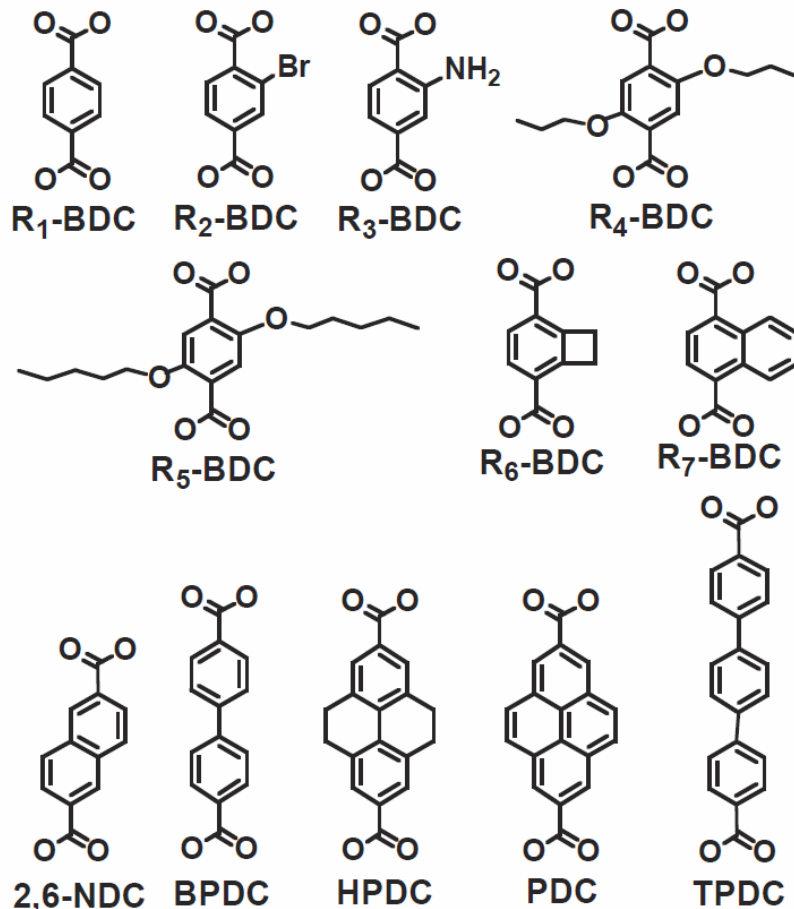
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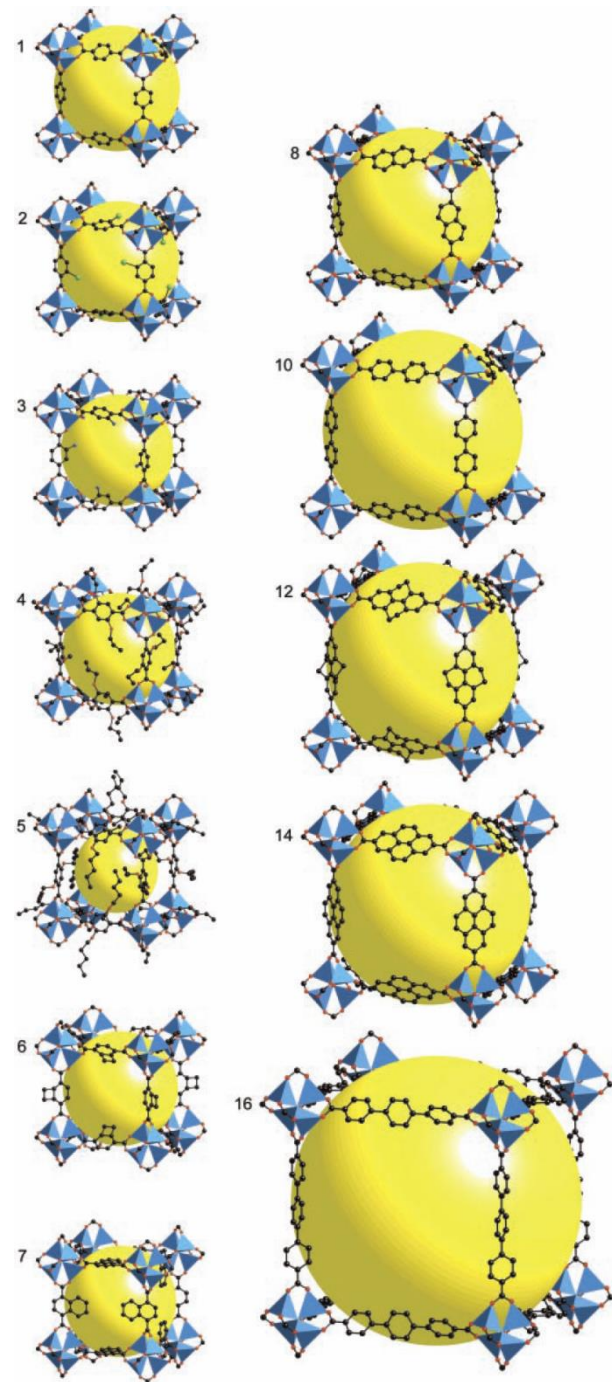
The concept of **isoreticular chemistry** was made popular in **2002** for a series of Zn dicarboxylates but has also been extended to other materials.

Isorecticular MOFs, IRMOFs



(IRMOF-1 = MOF-5)

M. Eddaoudi, J. Kim, N. Rosi, D. Vodak, J. Wachter, M. O'Keeffe and O. M. Yaghi, *Science*, **2002**, 295, 469-472.



MOF history

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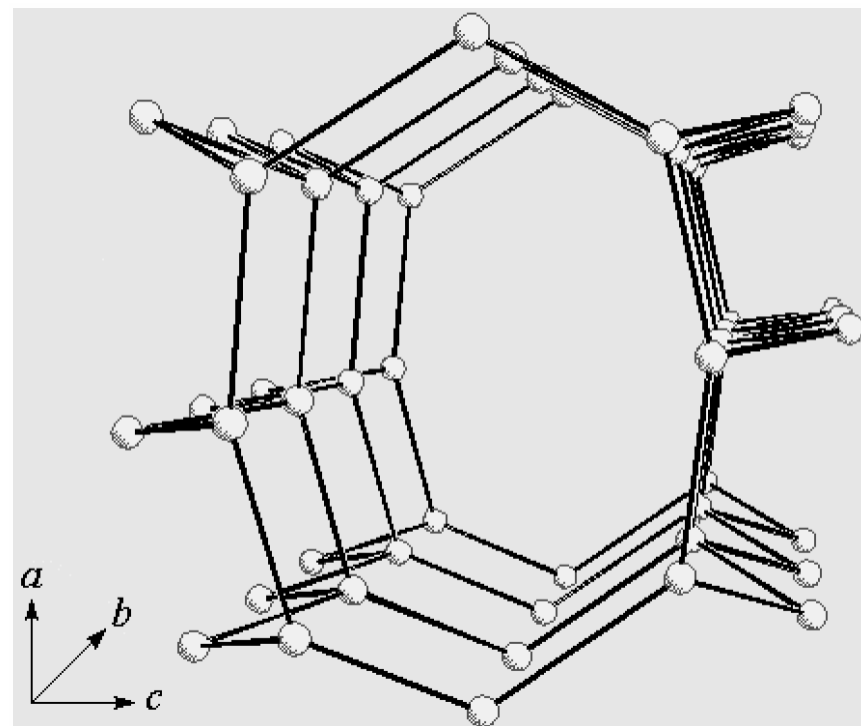
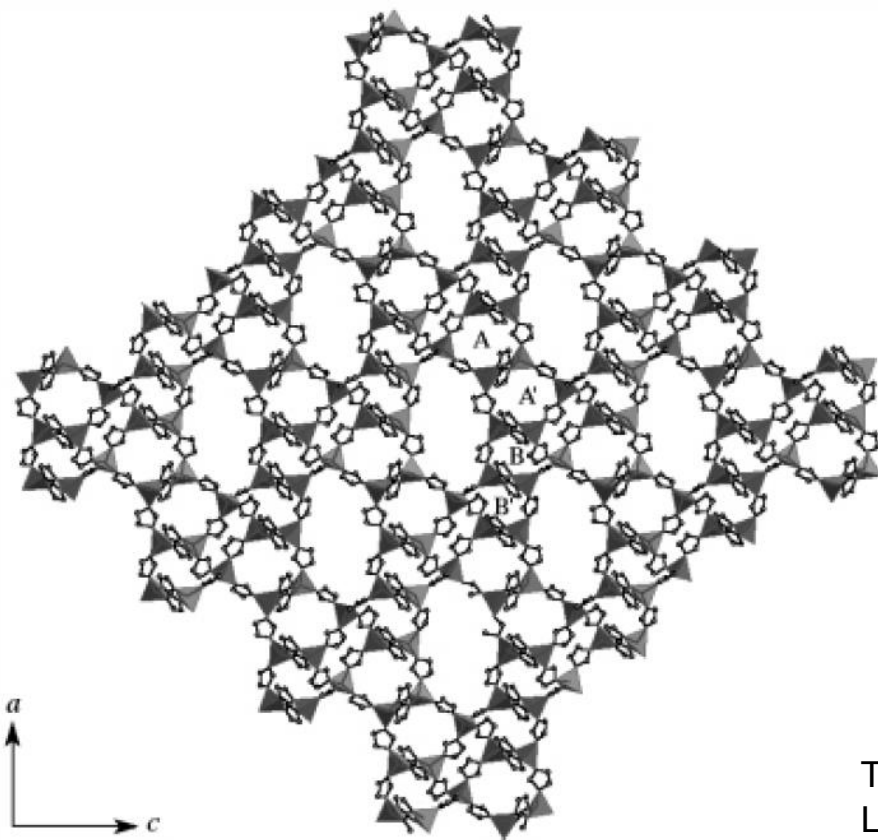
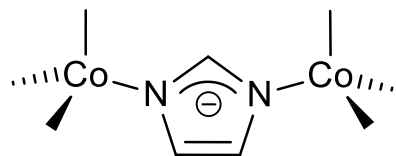
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The concept of **isoreticular chemistry** was made popular in **2002** for a series of Zn dicarboxylates but has also been extended to other materials.

In **2002** (more prominently noted in 2006), imidazolate-based compounds, nowadays known as zeolitic imidazolate frameworks (**ZIFs**), were prepared.

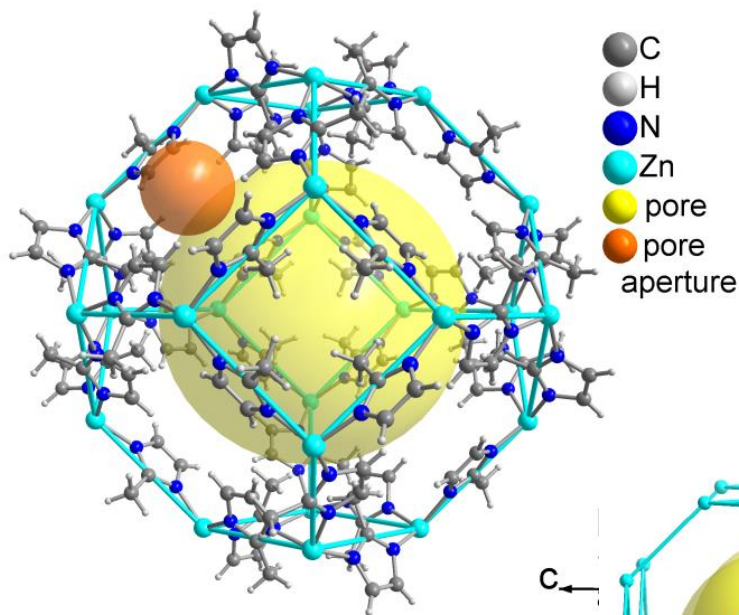
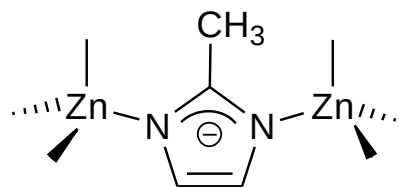
ZIF = zeolitic imidazolate framework
 $[\text{Co}_5(\text{im})_{10} \cdot 2\text{MB}]$

MB = 3-methyl-1-butanol



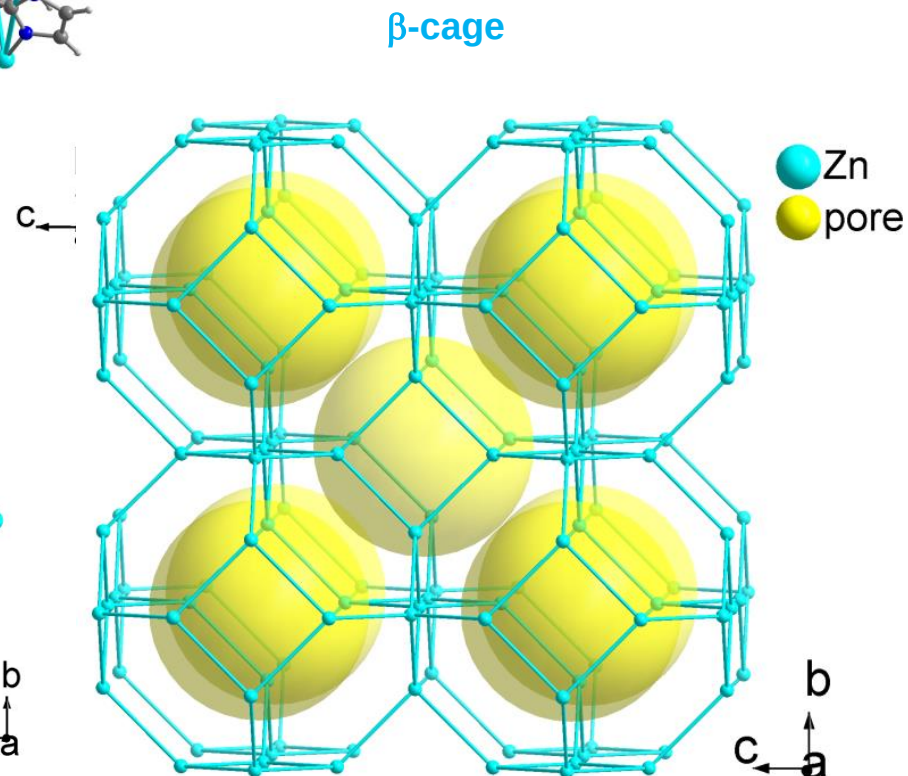
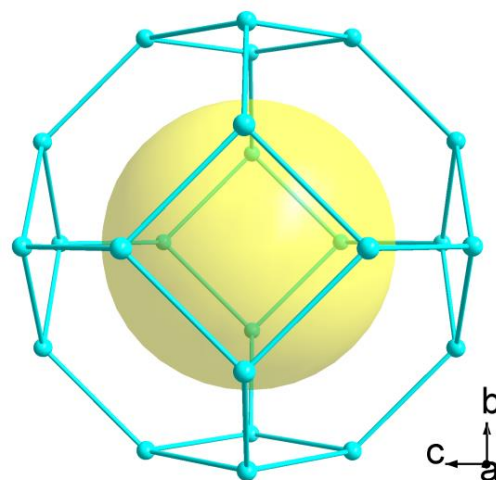
Tian, Y.-Q.; Cai, C.-X.; Ji, Y.; You, X.-Z.; Peng, S.-M.;
 Lee, G.-H. *Angew. Chem. Int. Ed.* **2002**, *41*, 1384-
 1386..

ZIF = zeolitic imidazolate framework
 ZIF-8 = 3D-[Zn-(mim)₂] · 2H₂O

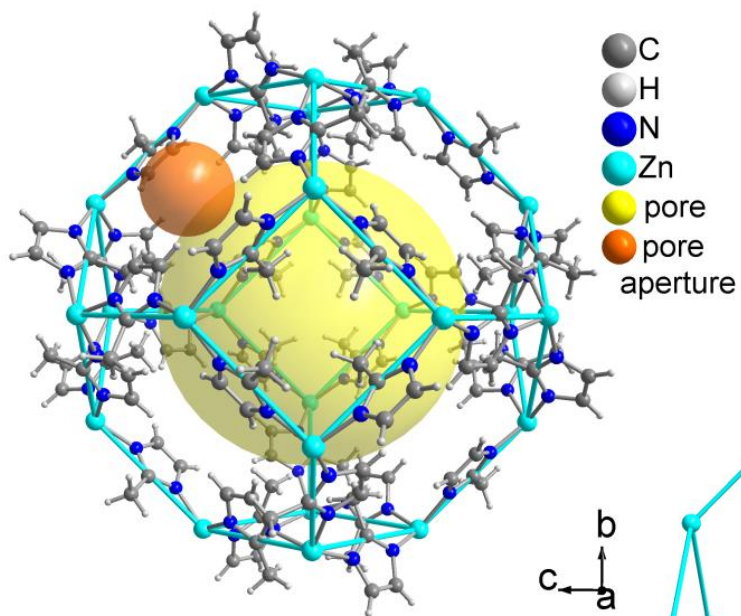
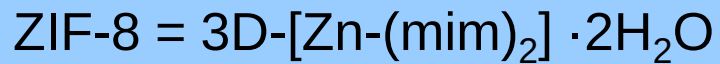


Huang, X.-C.; Lin, Y.-Y.; Zhang, J.-P.; Chen, X.-M. *Angew. Chem. Int. Ed.* **2006**, 45, 1557–1559.

Park, K. S.; Ni, Z.; Cote, A. P.; Choi, J. Y.; Huang, R.; Uribe-Romo, F. J.; Chae, H. K.; O'Keeffe, M.; Yaghi, O. M. *Proc. Natl. Acad. Sci. U.S.A.* **2006**, 103, 10186–10191

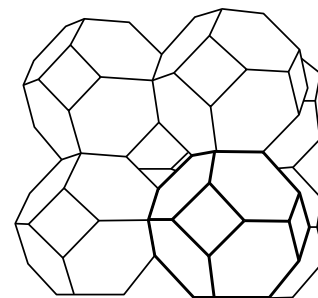
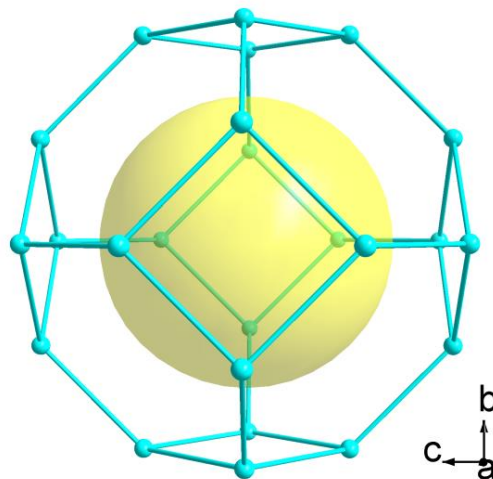


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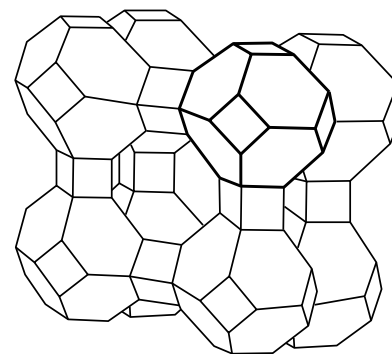


natural zeolites:

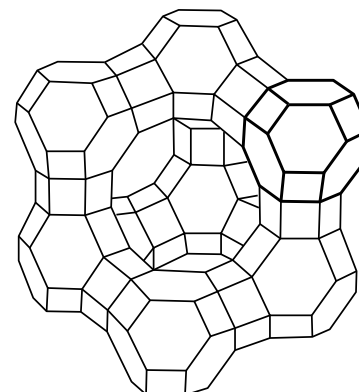
β -cage



sodalite

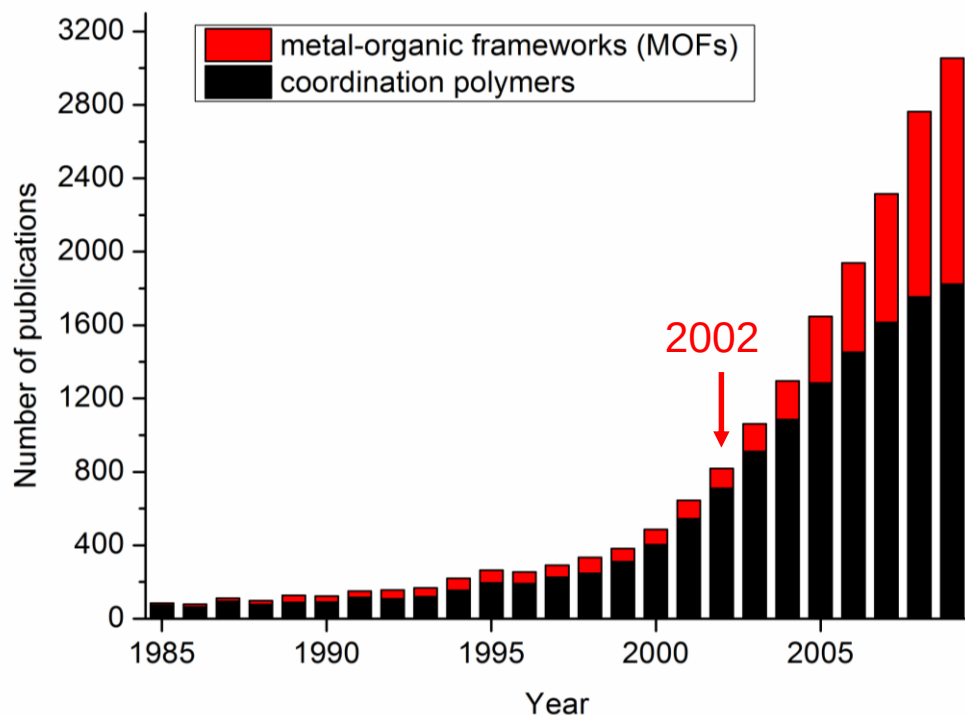


zeolite A



(c) faujasit

Coordination polymers / Metal-organic frameworks (MOFs): A dynamic field



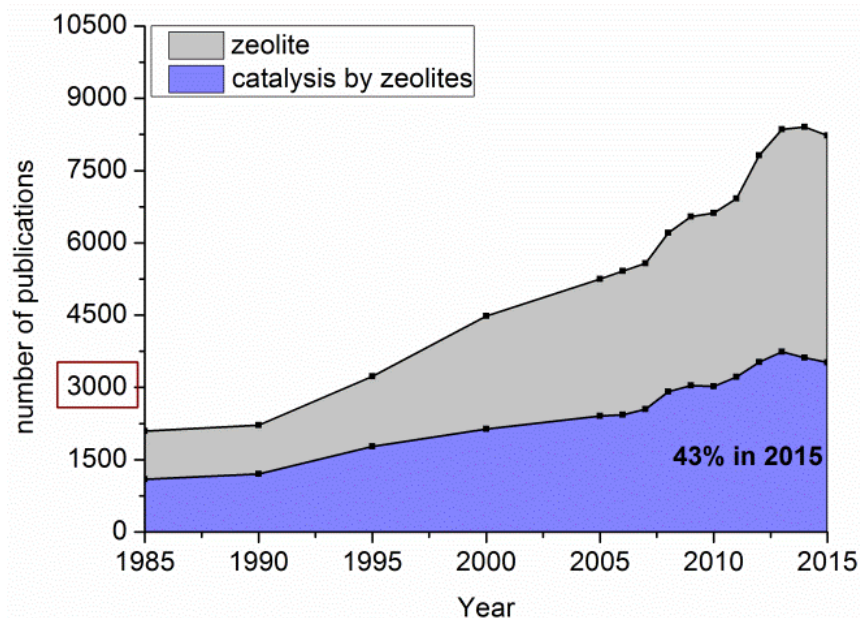
Scifinder: "metal organic framework", "MOF"
or "coordination polymer"

since 2002 the field of MOFs
developed with increasing pace ...

... but compared to zeolites ...

Metal-organic frameworks (MOFs): A dynamic field

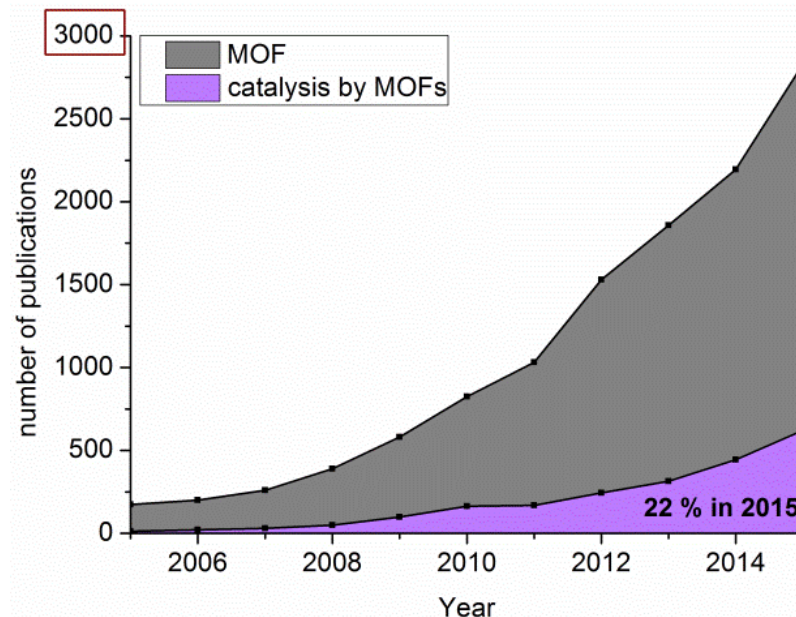
... but compared to zeolites ...



Scifinder "zeolites", "catalysis by zeolites"

MOFs still have to catch up!

Scifinder: "MOF", "catalysis by MOFs"



A. Herbst, C. Janiak,
CrystEngComm **2017**, 19, 4092-4117
DOI: 10.1039/C6CE01782G

Content

1. Definition and stability issue
2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look - new file
5. Synthesis and analysis
6. Selected applications